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(54) **AEROSOL GENERATING DEVICE**

(71) Applicant: **KT&G Corporation**, Daejeon (KR)

(72) Inventors: **Hyungjin JUNG**, Seoul (KR); **Taehun KIM**, Gyeonggi-do (KR); **Jueon PARK**, Seoul (KR); **Sungwook YOON**, Gyeonggi-do (KR); **Jungho HAN**, Daejeon (KR)

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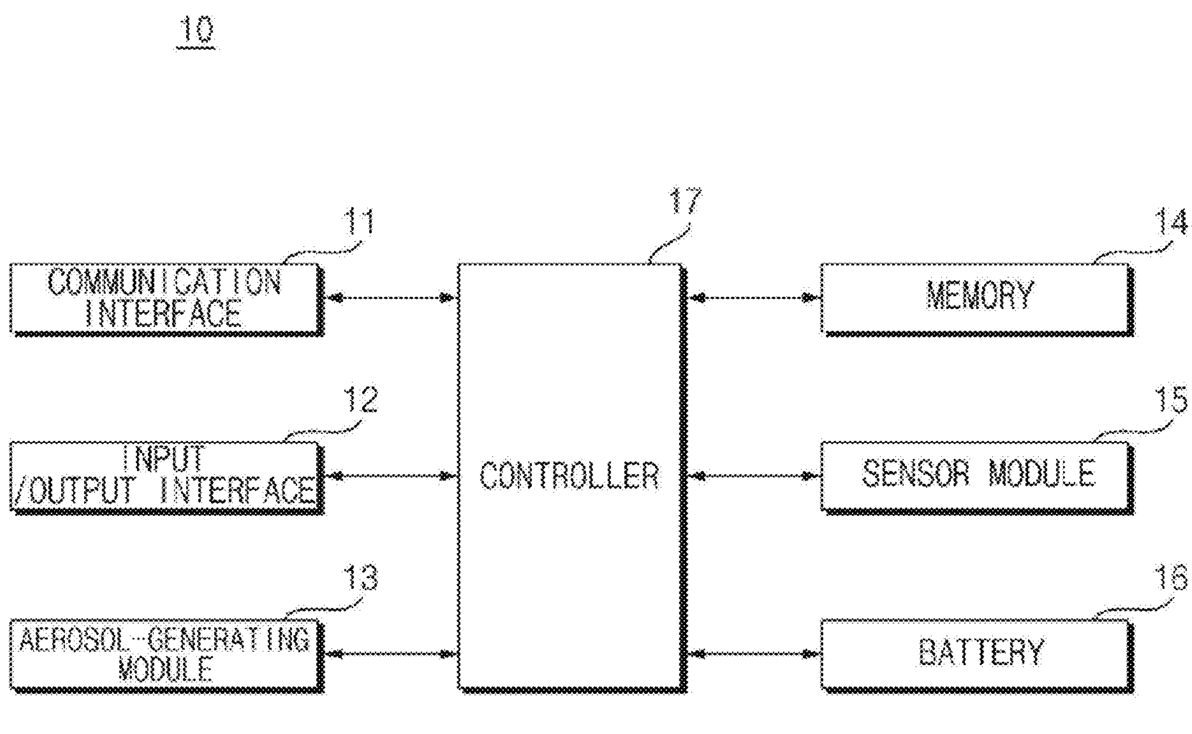
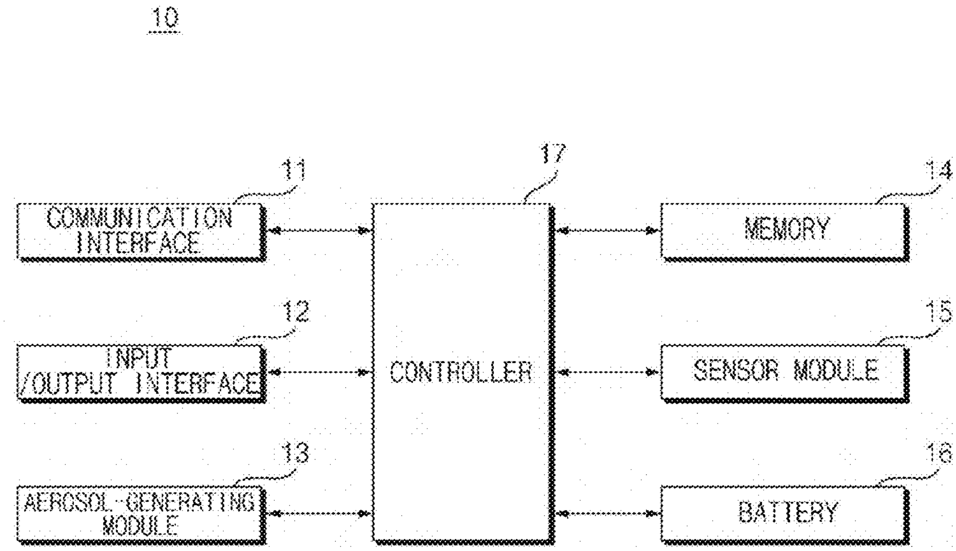
(57) **ABSTRACT**

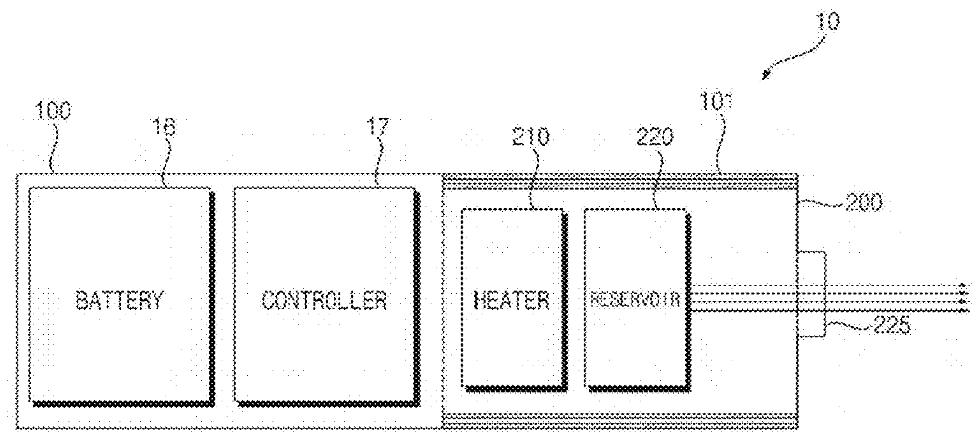
An aerosol-generating device is disclosed. The aerosol-generating device of the disclosure includes a chamber configured to store a liquid, a heater configured to heat the liquid. a resistance detection sensor configured to output a signal corresponding to a resistance of the heater, and a controller configured to calculate a temperature of the heater based on the resistance of the heater. The controller is configured to determine whether the temperature of the heater exceeds a first temperature in response to supply of sensing power to the heater in a first preheating period. determine whether the temperature of the heater exceeds a second temperature greater than the first temperature in response to supply of the sensing power to the heater in a second preheating period. based on the temperature of the heater exceeding the first temperature, and determine that the liquid is exhausted based on the temperature of the heater exceeding the second temperature.



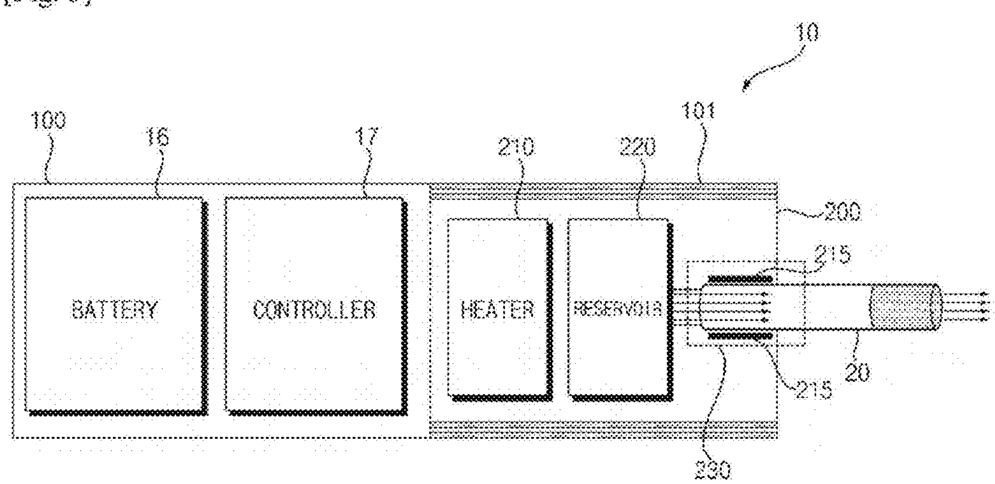
[Fig. 1]



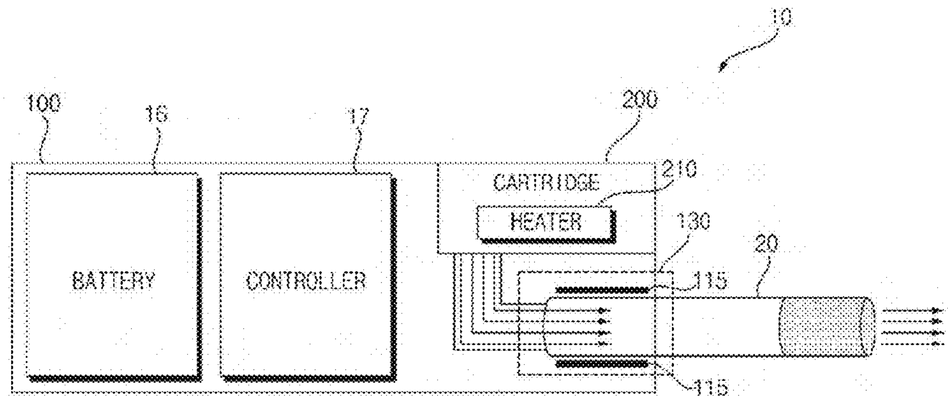
[Fig. 2]



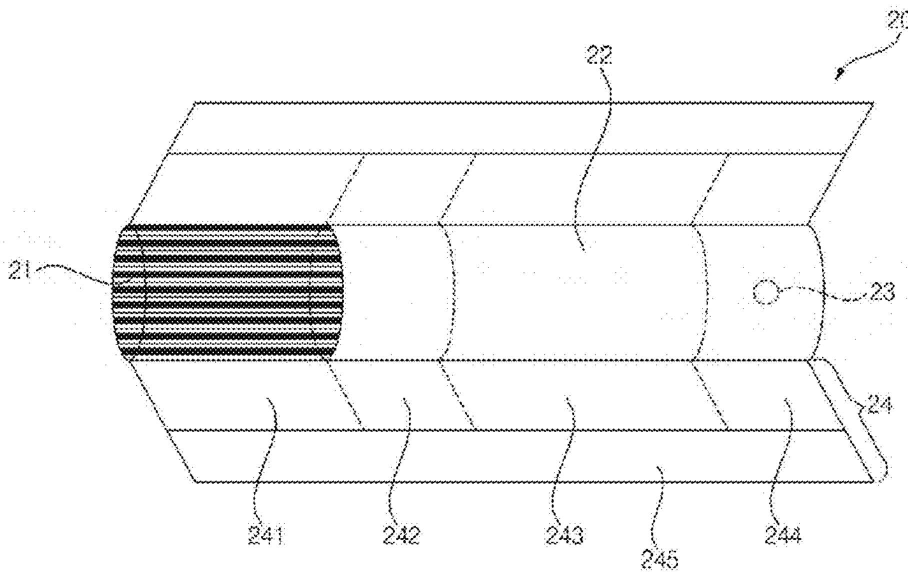
[Fig. 3]



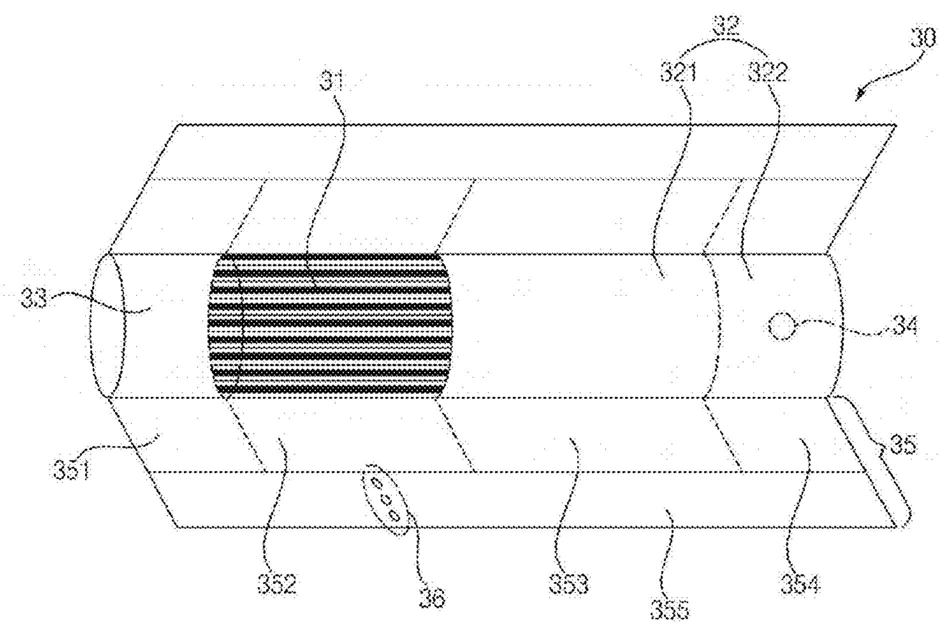
[Fig. 4]



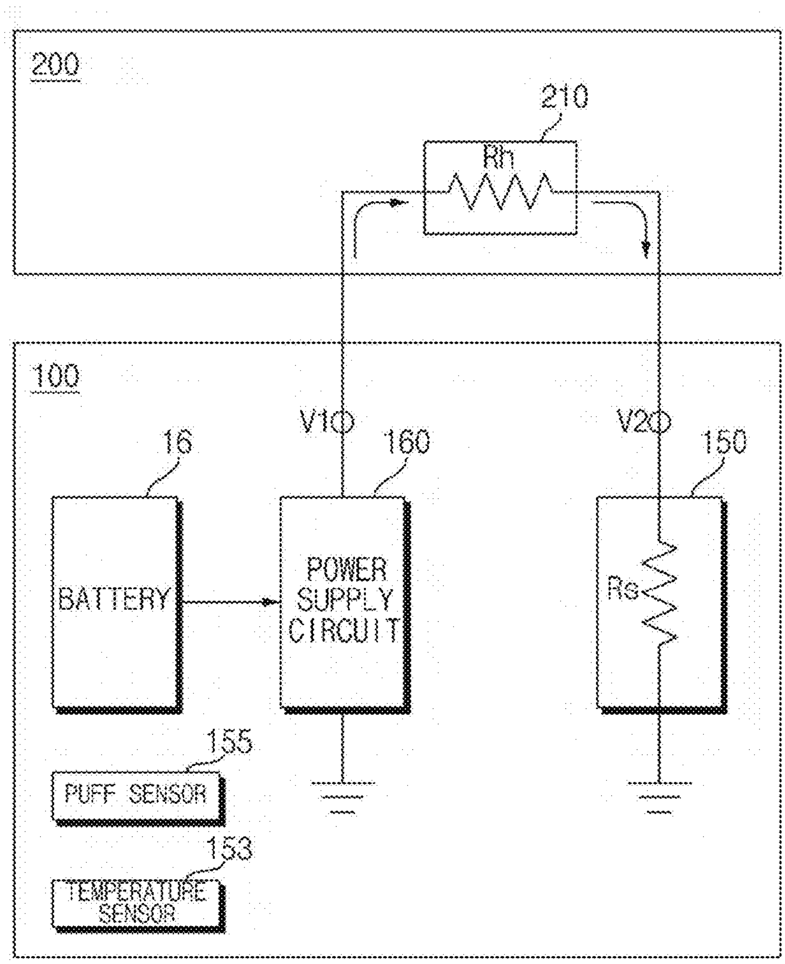
[Fig. 5]



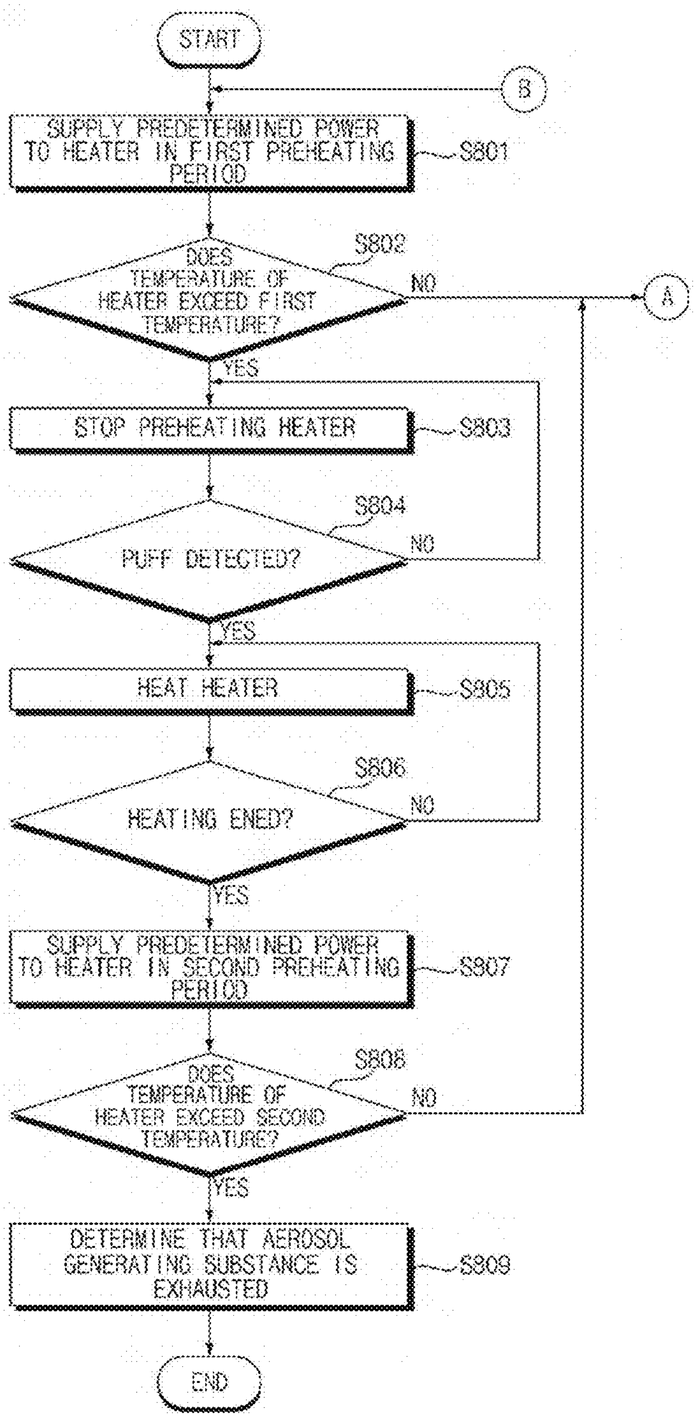
[Fig. 6]



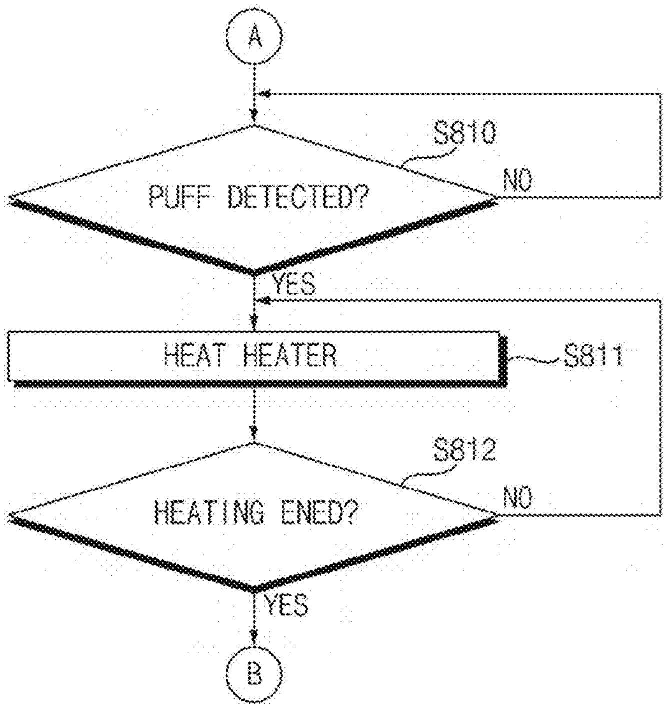
[Fig. 7]


200
210
Rh
100
V1
160
V2
150
16
BATTERY
POWER
SUPPLY
CIRCUIT
Rs
155
PUFF SENSOR
153
TEMPERATURE
SENSOR

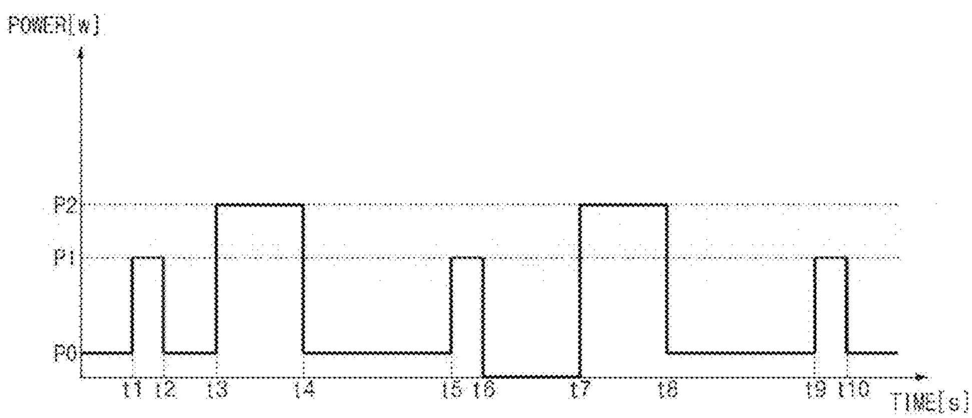
[Fig. 8a]


START
B
SUPPLY PREDETERMINED POWER TO HEATER IN FIRST PREHEATING PERIOD
S801
DOES TEMPERATURE OF HEATER EXCEED FIRST TEMPERATURE?
S802
NO
A
YES
STOP PREHEATING HEATER
S803
PUFF DETECTED?
S804
NO
YES
HEAT HEATER
S805
HEATING ENED?
S806
NO
YES
SUPPLY PREDETERMINED POWER TO HEATER IN SECOND PREHEATING PERIOD
S807
DOES TEMPERATURE OF HEATER EXCEED SECOND TEMPERATURE?
S808
NO
YES
DETERMINE THAT AEROSOL GENERATING SUBSTANCE IS EXHAUSTED
S809
END

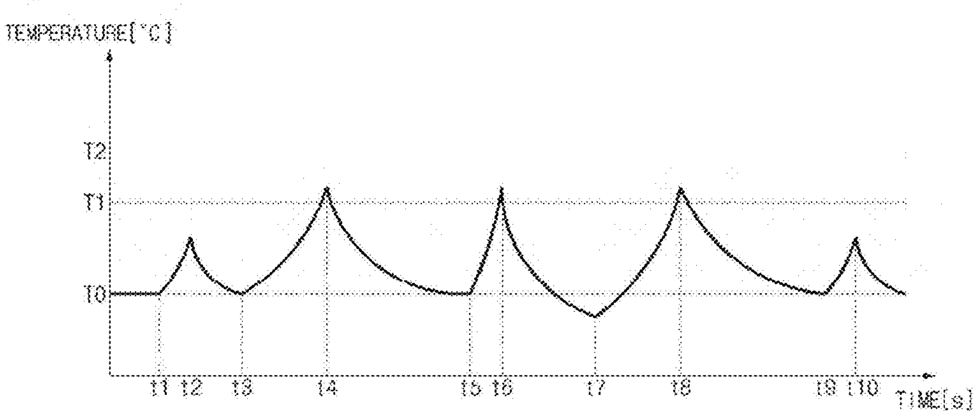
[Fig. 8b]



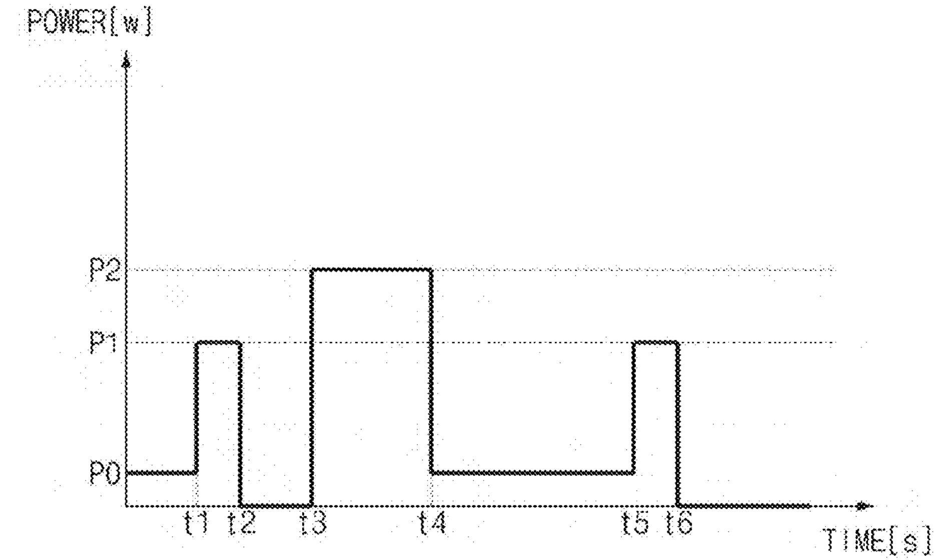
[Fig. 9]



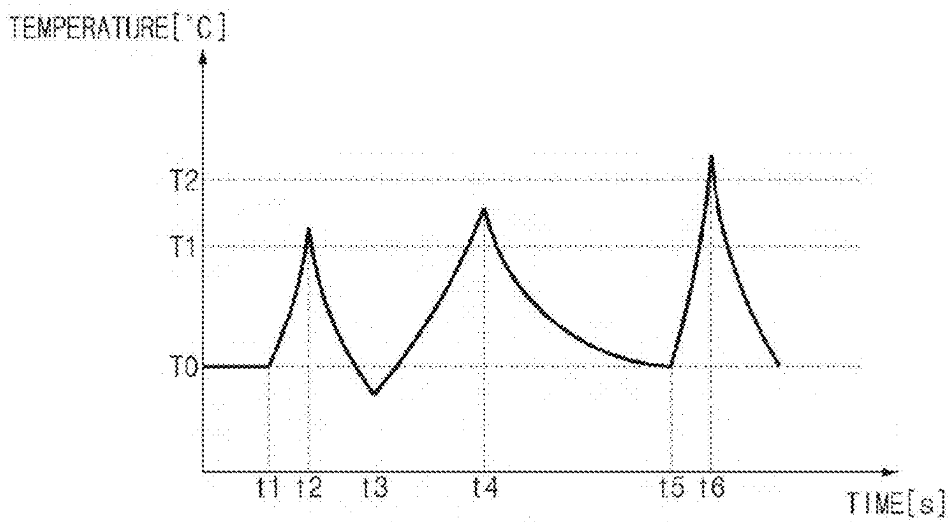
[Fig. 10]



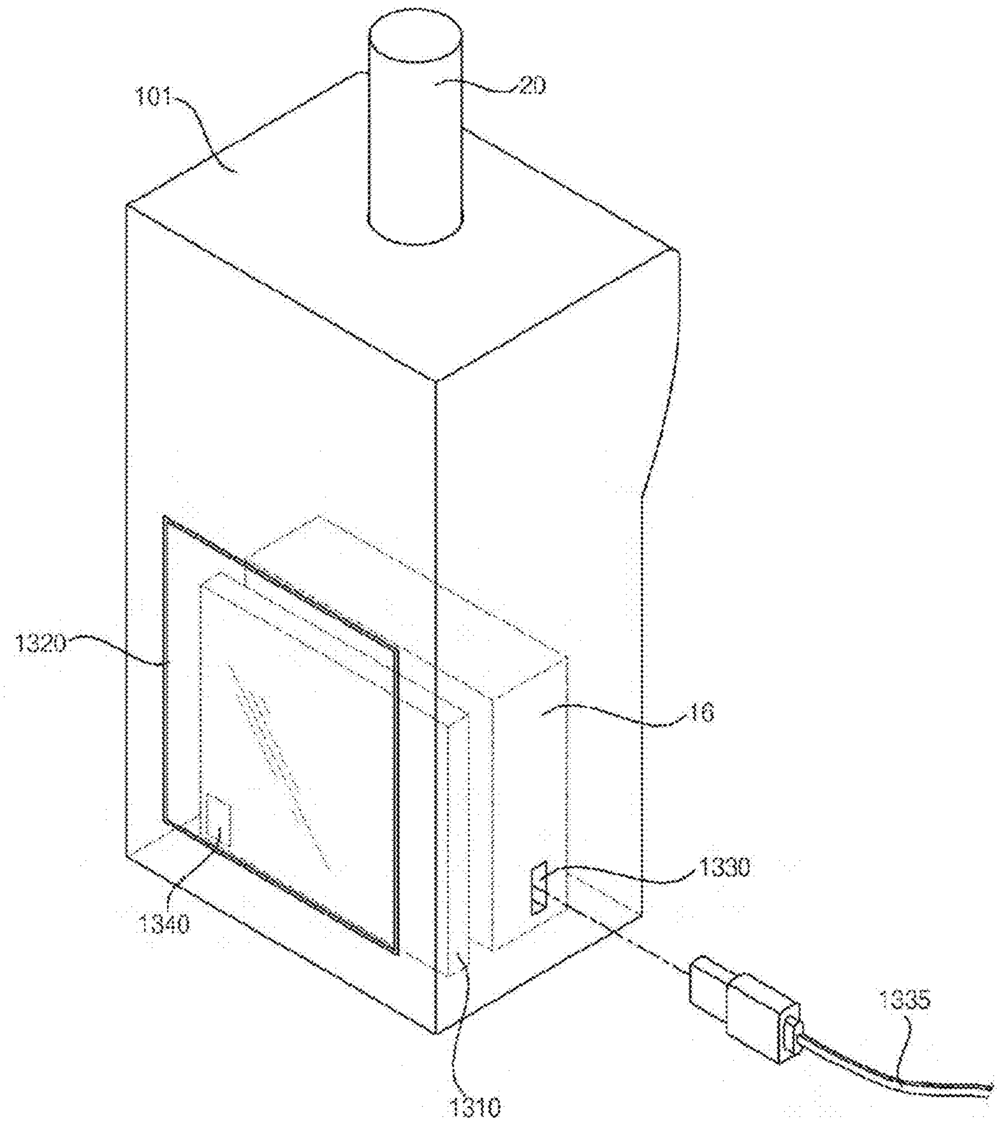
[Fig. 11]


POWER[w]
P2
P1
P0
t1 t2 t3 t4 t5 t6
TIME[s]

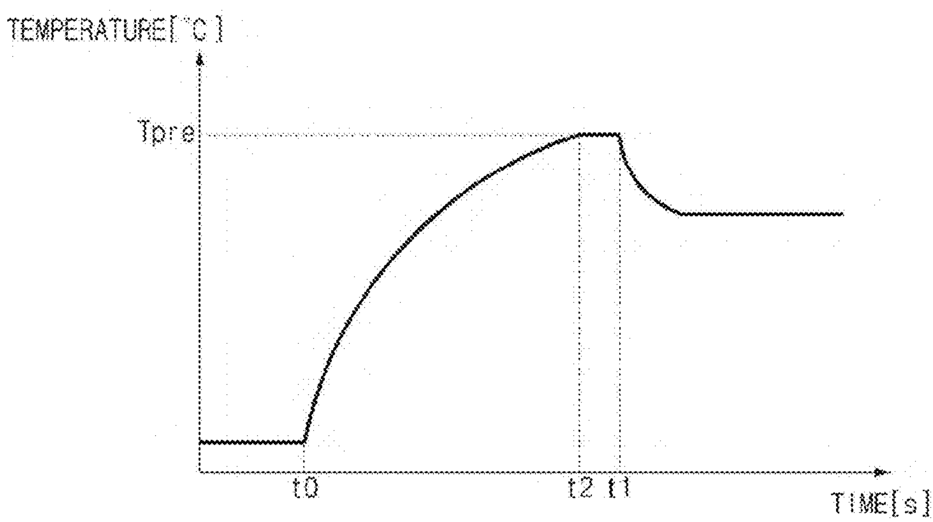
[Fig. 12]


TEMPERATURE[°C]
T2
T1
T0
t1 t2 t3 t4 t5 t6
TIME[s]

[Fig. 13]


101
20
1320
16
1330
1340
1335
1310

[Fig. 14]


TEMPERATURE[°C]
Tpre
t0
t2
t1
TIME[s]

AEROSOL GENERATING DEVICE

TECHNICAL FIELD

[0001] The present disclosure relates to an aerosol-generating device.

BACKGROUND ART

[0002] An aerosol-generating device is a device that extracts certain components from a medium or a substance by forming an aerosol. The medium may contain a multi-component substance. The substance contained in the medium may be a multi-component flavoring substance. For example, the substance contained in the medium may include a nicotine component, an herbal component, and/or a coffee component. Recently, various research on aerosol-generating devices has been conducted.

DISCLOSURE OF INVENTION

Technical Problem

[0003] It is an object of the present disclosure to solve the above and other problems.

[0004] It is another object of the present disclosure to provide an aerosol-generating device capable of determining whether a liquid aerosol-generating substance is smoothly supplied to a liquid delivery element based on a temperature of a heater in a preheating period.

[0005] It is still another object of the present disclosure to provide an aerosol-generating device capable of smoothly supplying a liquid aerosol-generating substance to a liquid delivery element when the liquid delivery clement lacks the liquid aerosol-generating substance.

[0006] It is still another object of the present disclosure to provide an aerosol-generating device capable of accurately determining whether a liquid aerosol-generating substance is exhausted based on a temperature of a heater in a preheating period.

Solution to Problem

[0007] An aerosol-generating device according to an aspect of the present disclosure for accomplishing the above and other objects may include a chamber configured to store a liquid, a heater configured to heat the liquid, a resistance detection sensor configured to output a signal corresponding to a resistance of the heater, and a controller configured to calculate a temperature of the heater based on the resistance of the heater. The controller is configured to determine whether the temperature of the heater exceeds a first temperature in response to supply of sensing power to the heater in a first preheating period, determine whether the temperature of the heater exceeds a second temperature greater than the first temperature in response to supply of the sensing power to the heater in a second preheating period, based on the temperature of the heater exceeding the first temperature, and determine that the liquid is exhausted based on the temperature of the heater exceeding the second temperature.

Advantageous Effects of Invention

[0008] According to at least one of embodiments of the present disclosure, it may be possible to determine whether a liquid aerosol-generating substance is smoothly supplied to a liquid delivery element based on a temperature of a heater in a preheating period.

[0009] According to at least one of embodiments of the present disclosure, it may be possible to smoothly supply a liquid aerosol-generating substance to a liquid delivery element when the liquid delivery element lacks the liquid aerosol-generating substance.

[0010] According to at least one of embodiments of the present disclosure, it may be possible to accurately determine whether a liquid aerosol-generating substance is exhausted based on a temperature of a heater in a preheating period.

[0011] Additional applications of the present disclosure will become apparent from the following detailed description. However, because various changes and modifications will be clearly understood by those skilled in the art within the spirit and scope of the present disclosure, it should be understood that the detailed description and specific embodiments, such as preferred embodiments of the present disclosure, are merely given by way of example.

BRIEF DESCRIPTION OF DRAWINGS

[0012] The above and other objects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0013] FIG. **1** is a block diagram of an aerosol-generating device according to an embodiment of the present disclosure;

[0014] FIGS. **2** to **4** are views for explaining an aerosol-generating device according to embodiments of the present disclosure;

[0015] FIGS. **5** and **6** are views for explaining a stick according to embodiments of the present disclosure;

[0016] FIG. **7** is a diagram for explaining the configuration of an aerosol-generating device according to an embodiment of the present disclosure;

[0017] FIGS. **8**A and **8**B are flowcharts showing an operation method of the aerosol-generating device according to an embodiment of the present disclosure; and

[0018] FIGS. **9** to **14** are diagrams for explaining the operation of an aerosol-generating device according to an embodiment of the present disclosure.

MODE FOR THE INVENTION

[0019] Hereinafter, the embodiments disclosed in the present specification will be described in detail with reference to the accompanying drawings. The same or similar elements are denoted by the same reference numerals even though they are depicted in different drawings, and redundant descriptions thereof will be omitted.

[0020] In the following description, with respect to constituent elements used in the following description, the suffixes “module” and “unit” are used only in consideration of facilitation of description. The “module” and “unit” are do not have mutually distinguished meanings or functions.

[0021] In addition, in the following description of the embodiments disclosed in the present specification, a detailed description of known functions and configurations incorporated herein will be omitted when the same may make the subject matter of the embodiments disclosed in the present specification rather unclear. In addition, the accompanying drawings are provided only for a better understanding of the embodiments disclosed in the present specification and are not intended to limit the technical ideas disclosed in the present specification. Therefore, it should be understood that the accompanying drawings include all modifications, equivalents, and substitutions within the scope and sprit of the present disclosure.

[0022] It will be understood that the terms "first", "second", etc., may be used herein to describe various components. However, these components should not be limited by these terms. These terms are only used to distinguish one component from another component.

[0023] It will be understood that when a component is referred to as being "connected to" or "coupled to" another component, it may be directly connected to or coupled to another component. However, it will be understood that intervening components may be present. On the other hand, when a component is referred to as being "directly connected to" or "directly coupled to" another component, there are no intervening components present.

[0024] As used herein, the singular form is intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0025] FIG. **1** is a block diagram of an aerosol-generating device according to an embodiment of the present disclosure.

[0026] Referring to FIG. **1**, an aerosol-generating device **10** may include a communication interface **11**, an input/output interface **12**, an aerosol-generating module **13**, a memory **14**, a sensor module **15**, a battery **16**, and/or a controller **17**.

[0027] In one embodiment, the aerosol-generating device **10** may be composed only of a main body. In this case, components included in the aerosol-generating device **10** may be located in the main body. In another embodiment, the aerosol-generating device **10** may be composed of a cartridge, which contains an aerosol-generating substance, and a main body. In this case, the components included in the aerosol-generating device **10** may be located in at least one of the main body or the cartridge.

[0028] The communication interface **11** may include at least one communication module for communication with an external device and/or a network. For example, the communication interface **11** may include a communication module for wired communication, such as a Universal Serial Bus (USB). For example, the communication interface **11** may include a communication module for wireless communication, such as Wireless Fidelity (Wi-Fi), Bluetooth, Bluetooth Low Energy (BLE), ZigBee, or nearfield communication (NFC).

[0029] The input/output interface **12** may include an input device (not shown) for receiving a command from a user and/or an output device (not shown) for outputting information to the user. For example, the input device may include a touch panel, a physical button, a microphone, or the like. For example, the output device may include a display device for outputting visual information, such as a display or a light-emitting diode (LED), an audio device for outputting auditory information, such as a speaker or a buzzer, a motor for outputting tactile information such as haptic effect, or the like.

[0030] The input/output interface **12** may transmit data corresponding to a command input by the user through the input device to another component (or other components) of the aerosol-generating device **100**. The input/output interface **12** may output information corresponding to data received from another component (or other components) of the aerosol-generating device **10** through the output device.

[0031] The aerosol-generating module **13** may generate an aerosol from an aerosol-generating substance. Here, the aerosol-generating substance may be a substance in a liquid state, a solid state, or a gel state, which is capable of generating an aerosol, or a combination of two or more aerosol-generating substances.

[0032] According to an embodiment, the liquid aerosol-generating substance may be a liquid including a tobacco-containing material having a volatile tobacco flavor component. According to another embodiment, the liquid aerosol-generating substance may be a liquid including a non-tobacco material. For example, the liquid aerosol-generating substance may include water, solvents, nicotine, plant extracts, flavorings, flavoring agents, vitamin mixtures, etc.

[0033] The solid aerosol-generating substance may include a solid material based on a tobacco raw material such as a reconstituted tobacco sheet, shredded tobacco, or granulated tobacco. In addition, the solid aerosol-generating substance may include a solid material having a taste control agent and a flavoring material. For example, the taste control agent may include calcium carbonate, sodium bicarbonate, calcium oxide, etc. For example, the flavoring material may include a natural material such as herbal granules, or may include a material such as silica, zeolite, or dextrin, which includes an aroma ingredient.

[0034] In addition, the aerosol-generating substance may further include an aerosol-forming agent such as glycerin or propylene glycol.

[0035] The aerosol-generating module **13** may include at least one heater (not shown).

[0036] The aerosol-generating module **13** may include an electro-resistive heater. For example, the electro-resistive heater may include at least one electrically conductive track. The electro-resistive heater may be heated as current flows through the electrically conductive track. At this time, the aerosol-generating substance may be heated by the heated electro-resistive heater.

[0037] The electrically conductive track may include an electro-resistive material. In one example, the electrically conductive track may be formed of a metal material. In another example, the electrically conductive track may be formed of a ceramic material, carbon, a metal alloy, or a composite of a ceramic material and metal.

[0038] The electro-resistive heater may include an electrically conductive track that is formed in any of various shapes. For example, the electrically conductive track may be formed in any one of a tubular shape, a plate shape, a needle shape, a rod shape, and a coil shape.

[0039] The aerosol-generating module **13** may include a heater that uses an induction-heating method. For example, the induction heater may include an electrically conductive coil. The induction heater may generate an alternating magnetic field, which periodically changes in direction, by adjusting the current flowing through the electrically conductive coil. At this time, when the alternating magnetic field is applied to a magnetic body, energy loss may occur in the magnetic body due to eddy current loss and hysteresis loss. In addition, the lost energy may be released as thermal energy. Accordingly, the aerosol-generating substance located adjacent to the magnetic body may be heated. Here, an object that generates heat due to the magnetic field may be referred to as a susceptor.

[0040] Meanwhile, the aerosol-generating module **13** may generate ultrasonic vibrations to thereby generate an aerosol from the aerosol-generating substance.

[0041] The aerosol-generating device **10** may be referred to as a cartomizer, an atomizer, or a vaporizer.

[0042] The memory **14** may store programs for processing and controlling each signal in the controller **17**. The memory **14** may store processed data and data to be processed.

[0043] For example, the memory **14** may store applications designed for the purpose of performing various tasks that can be processed by the controller **17**. The memory **14** may selectively provide some of the stored applications in response to the request from the controller **17**.

[0044] For example, the memory **14** may store data on the operation time of the aerosol-generating device **100**, the maximum number of puffs, the current number of puffs, the number of uses of battery **16**, at least one temperature profile, the user's inhalation pattern, and data about charging/discharging. Here, "puff" means inhalation by the user. "inhalation" means the user's act of taking air or other substances into the user's oral cavity, nasal cavity, or lungs through the user's mouth or nose.

[0045] The memory **14** may include at least one of volatile memory (e.g. dynamic random access memory (DRAM), static random access memory (SRAM), or synchronous dynamic random access memory (SDRAM)), nonvolatile memory (e.g. flash memory), a hard disk drive (HDD), or a solid-state drive (SSD).

[0046] The sensor module **15** may include at least one sensor.

[0047] For example, the sensor module **15** may include a sensor for sensing a puff (hereinafter referred to as a "puff sensor"). In this case, the puff sensor may be implemented as a proximity sensor such as an IR sensor, a pressure sensor, a gyro sensor, an acceleration sensor, a magnetic field sensor, or the like.

[0048] For example, the sensor module **15** may include a sensor for sensing a puff (hereinafter referred to as a "puff sensor"). In this case, the puff sensor may be implemented by a pressure sensor, a gyro sensor, an acceleration sensor, a magnetic field sensor, or the like.

[0049] For example, the sensor module **15** may include a sensor for sensing the temperature of the heater included in the aerosol-generating module **13** and the temperature of the aerosol-generating substance (hereinafter referred to as a "temperature sensor"). In this case, the heater included in the aerosol-generating module **13** may also serve as the temperature sensor. For example, the electro-resistive material of the heater may be a material having a predetermined temperature coefficient of resistance. The sensor module **15** may measure the resistance of the heater, which varies according to the temperature, to thereby sense the temperature of the heater.

[0050] For example, in the case in which the main body of the aerosol-generating device **10** is formed to allow a stick to be inserted thereinto, the sensor module **15** may include a sensor for sensing insertion of the stick (hereinafter referred to as a "stick detection sensor").

[0051] For example, in the case in which the aerosol-generating device **10** includes a cartridge, the sensor module **15** may include a sensor for sensing mounting/demounting of the cartridge and the position of the cartridge (hereinafter referred to as a "cartridge detection sensor").

[0052] In this case, the stick detection sensor and/or the cartridge detection sensor may be implemented as an inductance-based sensor, a capacitive sensor, a resistance sensor, or a Hall sensor (or Hall IC) using a Hall effect.

[0053] For example, the sensor module **15** may include a voltage sensor for sensing a voltage applied to a component (e.g. the battery **16**) provided in the aerosol-generating device **10** and/or a current sensor for sensing a current.

[0054] The battery **16** may supply electric power used for the operation of the aerosol-generating device **10** under the control of the controller **17**. The battery **16** may supply electric power to other components provided in the aerosol-generating device **100**. For example, the battery **16** may supply electric power to the communication module included in the communication interface **11**, the output device included in the input/output interface **12**, and the heater included in the aerosol-generating module **13**.

[0055] The battery **16** may be a rechargeable battery or a disposable battery. For example, the battery **16** may be a lithium-ion (Li-ion) battery or a lithium polymer (Li-polymer) battery. However, the present disclosure is not limited thereto. For example, when the battery **16** is rechargeable, the charging rate (C-rate) of the battery **16** may be 10 C, and the discharging rate (C-rate) thereof may be 10 C to 20 C. However, the present disclosure is not limited thereto. Also, for stable use, the battery **16** may be manufactured such that 80% or more of the total capacity may be ensured even when charging/discharging is performed 2000 times.

[0056] The aerosol-generating device **10** may further include a protection circuit module (PCM) (not shown), which is a circuit for protecting the battery **16**. The protection circuit module (PCM) may be disposed adjacent to the upper surface of the battery **16**. For example, in order to prevent overcharging and overdischarging of the battery **16**, the protection circuit module (PCM) may cut off the electrical path to the battery **16** when a short circuit occurs in a circuit connected to the battery **16**, when an overvoltage is applied to the battery **16**, or when an overcurrent flows through the battery **16**.

[0057] The aerosol-generating device **10** may further include a charging terminal to which electric power supplied from the outside is input. For example, the charging terminal may be formed at one side of the main body of the aerosol-generating device **100**. The aerosol-generating device **10** may charge the battery **16** using electric power supplied through the charging terminal. In this case, the charging terminal may be configured as a wired terminal for USB communication, a pogo pin, or the like.

[0058] The aerosol-generating device **10** may wirelessly receive electric power supplied from the outside through the communication interface **11**. For example, the aerosol-generating device **10** may wirelessly receive electric power using an antenna included in the communication module for wireless communication. The aerosol-generating device **10** may charge the battery **16** using the wirelessly supplied electric power.

[0059] The controller **17** may control the overall operation of the aerosol-generating device **100**. The controller **17** may be connected to each of the components provided in the aerosol-generating device **100**. The controller **17** may transmit and/or receive a signal to and/or from each of the components, thereby controlling the overall operation of each of the components.

[0060] The controller **17** may include at least one processor. The controller **17** may control the overall operation of the aerosol-generating device **10** using the processor included therein. Here, the processor may be a general processor such as a central processing unit (CPU). Of course, the processor may be a dedicated device such as an application-specific integrated circuit (ASIC), or may be any of other hardware-based processors.

[0061] The controller **17** may perform any one of a plurality of functions of the aerosol-generating device **100**. For example, the controller **17** may perform any one of a plurality of functions of the aerosol-generating device **10** (e.g. a preheating function, a heating function, a charging function, and a cleaning function) according to the state of each of the components provided in the aerosol-generating device **10** and the user's command received through the input/output interface **12**.

[0062] The controller **17** may control the operation of each of the components provided in the aerosol-generating device **10** based on data stored in the memory **14**. For example, the controller **17** may control the supply of a predetermined amount of electric power from the battery **16** to the aerosol-generating module **13** for a predetermined time based on the data on the temperature profile, the user's inhalation pattern, which is stored in the memory **14**.

[0063] The controller **17** may determine the occurrence or non-occurrence of a puff using the puff sensor included in the sensor module **15**. For example, the controller **17** may check a temperature change, a flow change, a pressure change, and a voltage change in the aerosol-generating device **10** based on the values sensed by the puff sensor. The controller **17** may determine the occurrence or non-occurrence of a puff based on the value sensed by the puff sensor.

[0064] The controller **17** may control the operation of each of the components provided in the aerosol-generating device **10** according to the occurrence or non-occurrence of a puff and/or the number of puffs. For example, the controller **17** may perform control such that the temperature of the heater is changed or maintained based on the temperature profile stored in the memory **14**.

[0065] The controller **17** may perform control such that the supply of electric power to the heater is interrupted according to a predetermined condition. For example, the controller **17** may perform control such that the supply of electric power to the heater is interrupted when the stick is removed, when the cartridge is demounted, when the number of puffs reaches the predetermined maximum number of puffs, when a puff is not sensed during a predetermined period of time or longer, or when the remaining capacity of the battery **16** is less than a predetermined value.

[0066] The controller **17** may calculate the remaining capacity with respect to the full charge capacity of the battery **16**. For example, the controller **17** may calculate the remaining capacity of the battery **16** based on the values sensed by the voltage sensor and/or the current sensor included in the sensor module **15**.

[0067] The controller **17** may perform control such that electric power is supplied to the heater using at least one of a pulse width modulation (PWM) method or a proportional-integral-differential (PID) method.

[0068] For example, the controller **17** may perform control such that a current pulse having a predetermined frequency and a predetermined duty ratio is supplied to the heater using the PWM method. In this case, the controller **17** may control the amount of electric power supplied to the heater by adjusting the frequency and the duty ratio of the current pulse.

[0069] For example, the controller **17** may determine a target temperature to be controlled based on the temperature profile. In this case, the controller **17** may control the amount of electric power supplied to the heater using the PID method, which is a feedback control method using a difference value between the temperature of the heater and the target temperature, a value obtained by integrating the difference value with respect to time, and a value obtained by differentiating the difference value with respect to time.

[0070] Although the PWM method and the PID method are described as examples of methods of controlling the supply of electric power to the heater, the present disclosure is not limited thereto, and may employ any of various control methods, such as a pro-portional-integral (PI) method or a proportional-differential (PD) method.

[0071] Meanwhile, the controller **17** may perform control such that electric power is supplied to the heater according to a predetermined condition. For example, when a cleaning function for cleaning the space into which the stick is inserted is selected in response to a command input by the user through the input/output interface **12**, the controller **17** may perform control such that a predetermined amount of electric power is supplied to the heater.

[0072] FIGS. **2** to **4** are views for explaining an aerosol-generating device according to embodiments of the present disclosure.

[0073] According to various embodiments of the present disclosure, the aerosol-generating device **10** may include a main body **100** and/or a cartridge **200**.

[0074] Referring to FIG. **2**, the aerosol-generating device **10** according to an embodiment may include a main body **100** and a cartridge **200**. The main body **100** may support the cartridge **200**, and the cartridge **200** may contain an aerosol-generating substance.

[0075] According to one embodiment, the cartridge **200** may be configured so as to be detachably mounted to the main body **100**. According to another embodiment, the cartridge **200** may be integrally configured with the main body **100**. For example, the cartridge **200** may be mounted to the main body **100** in a manner such that at least a portion of the cartridge **200** is inserted into the insertion space formed by a housing **101** of the main body **100**.

[0076] The main body **100** may be formed to have a structure in which external air can be introduced into the main body **100** in the state in which the cartridge **200** is inserted thereinto. Here, the external air introduced into the main body **100** may flow into the user's mouth via the cartridge **200**.

[0077] The controller **17** may determine whether the cartridge **200** is in a mounted state or a detached state using a cartridge detection sensor included in the sensor module **15**. For example, the cartridge detection sensor may transmit a pulse current through a first terminal connected with the cartridge **200**. In this case, the controller **17** may determine whether the cartridge **200** is in a connected state, based on whether the pulse current is received through a second terminal.

[0078] The cartridge **200** may include a reservoir **220** configured to contain the aerosol-generating substance and/or a heater **210** configured to heat the aerosol-generating substance in the reservoir **220**. For example, a liquid delivery element impregnated with (containing) the aerosol-generating substance may be disposed inside the reservoir **220**. The electrically conductive track of the heater **210** may be formed in a structure that is wound around the liquid delivery element. In this case, when the liquid delivery element is heated by the heater **210**, an aerosol may be generated. Here, the liquid delivery element may include a wick made of, for example, cotton fiber, ceramic fiber, glass fiber, or porous ceramic.

[0079] The cartridge **200** may include a mouthpiece **225**. Here, the mouthpiece **225** may be a portion to be inserted into a user's oral cavity. The mouthpiece **225** may have a discharge hole through which the aerosol is discharged to the outside during a puff.

[0080] Referring to FIG. **3**, the cartridge **200** may include an insertion space **230** configured to allow a stick **20** to be inserted. For example, the cartridge **200** may include the insertion space formed by an inner wall extending in a circumferential direction along a direction in which the stick **20** is inserted. In this case, the insertion space may be formed by opening the inner side of the inner wall up and down. The stick **20** may be inserted into the insertion space formed by the inner wall.

[0081] The insertion space into which the stick **20** is inserted may be formed in a shape corresponding to the shape of a portion of the stick **20** inserted into the insertion space. For example, when the stick **20** is formed in a cylindrical shape, the insertion space may be formed in a cylindrical shape.

[0082] When the stick **20** is inserted into the insertion space, the outer surface of the stick **20** may be surrounded by the inner wall and contact the inner wall.

[0083] A portion of the stick **20** may be inserted into the insertion space, the remaining portion of the stick **20** may be exposed to the outside.

[0084] The user may inhale the aerosol while biting one end of the stick **20** with the mouth. The aerosol generated by the heater **210** may pass through the stick **20** and be delivered to the user's mouth. At this time, while the aerosol passes through the stick **20**, the material contained in the stick **20** may be added to the aerosol. The material-infused aerosol may be inhaled into the user's oral cavity through the one end of the stick **20**.

[0085] The controller **17** may monitor the number of puffs based on the value sensed by the puff sensor from the point in time at which the stick **20** was inserted.

[0086] When the stick **20** is removed, the controller **17** may initialize the current number of puffs stored in the memory **14**.

[0087] The cartridge **200** may include a second heater **215** configured to heat the stick **20**. The second heater **215** may be disposed in the cartridge **200** at a position corresponding to a position at which the stick **20** is located after being inserted into the insertion space **230**. The second heater **215** may be implemented as an electrically conductive heater and/or an induction heating type heater. The second heater **215** may heat the inside and/or the outside of the stick **20** using the power supplied from the battery **16**.

[0088] Referring to FIG. **4**, the aerosol-generating device **10** according to an embodiment may include a main body **100** supporting the cartridge **200** and a cartridge **200** containing an aerosol-generating substance. The main body **100** may be formed so as to allow the stick **20** to be inserted into an insertion space **1300** therein.

[0089] The aerosol-generating device **10** may include the first heater **210** for heating for heating the aerosol-generating substance stored in the cartridge **200** and/or the second heater **215** for heating the stick **20** inserted into the main body **100**. For example, the aerosol-generating device **10** may generate an aerosol by heating the aerosol-generating substance stored in the cartridge **200** and the stick **20** using the first heater **210** and the second heater **115**, respectively.

[0090] The stick **20** may be similar to a general combustive cigarette. For example, the stick **20** may be divided into a first portion including an aerosol generating material and a second portion including a filter and the like. Alternatively, an aerosol generating material may be included in the second portion of the stick **20**. For example, a flavoring substance made in the form of granules or capsules may be inserted into the second portion.

[0091] Hereinafter, the present disclosure will be described on the basis of an embodiment in which the stick **20** is inserted into the insertion space **130** defined in the housing **101** of the main body **100**.

[0092] FIGS. **5** and **6** are views for explaining a stick according to embodiments of the present disclosure.

[0093] Referring to FIG. **5**, the stick **20** may include a tobacco rod **21** and a filter rod **22**. The first portion described above with reference to FIG. **4** may include the tobacco rod. The second portion described above with reference to FIG. **4** may include the filter rod **22**.

[0094] FIG. **5** illustrates that the filter rod **22** includes a single segment. However, the filter rod **22** is not limited thereto. In other words, the filter rod **22** may include a plurality of segments. For example, the filter rod **22** may include a first segment configured to cool an aerosol and a second segment configured to filter a certain component included in the aerosol. Also, as necessary, the filter rod **22** may further include at least one segment configured to perform other functions.

[0095] A diameter of the stick **20** may be within a range of 5 mm to 9 mm, and a length of the stick **20** may be about 48 mm, but embodiments are not limited thereto. For example, a length of the tobacco rod **21** may be about 12 mm, a length of a first segment of the filter rod **22** may be about 10 mm, a length of a second segment of the filter rod **22** may be about 14 mm, and a length of a third segment of the filter rod **22** may be about 12 mm, but embodiments are not limited thereto.

[0096] The stick **20** may be wrapped using at least one wrapper **24**. The wrapper **24** may have at least one hole through which external air may be introduced or internal air may be discharged. For example, the stick **20** may be wrapped using one wrapper **24**. As another example, the stick **20** may be double-wrapped using at least two wrappers **24**. For example, the tobacco rod **21** may be wrapped using a first wrapper **241**. For example, the filter rod **22** may be wrapped using wrappers **242**, **243**, **244**. The tobacco rod **21** and the filter rod **22** wrapped by wrappers may be combined. The stick **20** may be re-wrapped by a single wrapper **245**. When each of the tobacco rod **21** and the filter rod **22** includes a plurality of segments, each segment may be wrapped using wrappers **242**, **243**, **244**. The entirety of stick

20 composed of a plurality of segments wrapped by wrappers may be re-wrapped by another wrapper

[0097] The first wrapper **241** and the second wrapper **242** may be formed of general filter wrapping paper. For example, the first wrapper **241** and the second wrapper **242** may be porous wrapping paper or non-porous wrapping paper. Also, the first wrapper **241** and the second wrapper **242** may be made of an oil-resistant paper sheet and an aluminum laminate packaging material.

[0098] The third wrapper **243** may be made of a hard wrapping paper. For example, a basis weight of the third wrapper **243** may be within a range of 88 g/m2 to 96 g/m2. For example, the basis weight of the third wrapper **243** may be within a range of 90 g/m2 to 94 g/m2. Also, a total thickness of the third wrapper **243** may be within a range of 1200 μm to 1300 μm. For example, the total thickness of the third wrapper **243** may be 125 μm.

[0099] The fourth wrapper **244** may be made of an oil-resistant hard wrapping paper. For example, a basis weight of the fourth wrapper **244** may be within a range of about 88g/m2 to about 96 g/m2. For example, the basis weight of the fourth wrapper **244** may be within a range of 90 g/m2 to 94 g/m2. Also, a total thickness of the fourth wrapper **244** may be within a range of 1200 μm to 1300 μm. For example, the total thickness of the fourth wrapper **244** may be 125 μm.

[0100] The fifth wrapper **245** may be made of a sterilized paper (MFW). Here, the MFW refers to a paper specially manufactured to have enhanced tensile strength, water resistance, smoothness, and the like, compared to ordinary paper. For example, a basis weight of the fifth wrapper **245** may be within a range of 57 g/m2 to 63 g/m2. For example, a basis weight of the fifth wrapper **245** may be about 60 g/m2. Also, the total thickness of the fifth wrapper **245** may be within a range of 64 μm to 70 μm. For example, the total thickness of the fifth wrapper **245** may be 67 μm.

[0101] A predetermined material may be included in the fifth wrapper **245**. Here, an example of the predetermined material may be, but is not limited to, silicon. For example, silicon exhibits characteristics like heat resistance with little change due to the temperature, oxidation resistance, resistances to various chemicals, water repellency, electrical insulation, etc. However, any material other than silicon may be applied to (or coated on) the fifth wrapper **245** without limitation as long as the material has the above-mentioned characteristics.

[0102] The fifth wrapper **245** may prevent the stick **20** from being burned. For example, when the tobacco rod **21** is heated by the heater **110**, there is a possibility that the stick **20** is burned. In detail, when the temperature is raised to a temperature above the ignition point of any one of materials included in the tobacco rod **21**, the stick **20** may be burned. Even in this case, since the fifth wrapper **245** include a non-combustible material, the burning of the stick **20** may be prevented.

[0103] Furthermore, the fifth wrapper **245** may prevent the aerosol generating device **100** from being contaminated by substances formed by the stick **20**. Through puffs of a user, liquid substances may be formed in the stick **20**. For example, as the aerosol formed by the stick **20** is cooled by the outside air, liquid materials (e.g., moisture, etc.) may be formed. As the fifth wrapper **245** wraps the stick **20**, the liquid materials formed in the stick **20** may be prevented from being leaked out of the stick **20**.

[0104] The tobacco rod **21** may include an aerosol generating material. For example, the aerosol generating material may include at least one of glycerin, propylene glycol, ethylene glycol, dipropylene glycol, diethylene glycol, triethylene glycol, tetracthylene glycol, and oleyl alcohol, but it is not limited thereto. Also, the tobacco rod **21** may include other additives, such as flavors, a wetting agent, and/or organic acid. Also, the tobacco rod **21** may include a flavored liquid, such as menthol or a moisturizer, which is injected to the tobacco rod **21**.

[0105] The tobacco rod **21** may be manufactured in various forms. For example, the tobacco rod **21** may be formed as a sheet or a strand. Also, the tobacco rod **21** may be formed as a pipe tobacco, which is formed of tiny bits cut from a tobacco sheet. Also, the tobacco rod **21** may be surrounded by a heat conductive material. For example, the heat-conducting material may be, but is not limited to, a metal foil such as aluminum foil. For example, the heat conductive material surrounding the tobacco rod **21** may uniformly distribute heat transmitted to the tobacco rod **21**, and thus, the heat conductivity applied to the tobacco rod may be increased and taste of the tobacco may be improved. Also, the heat conductive material surrounding the tobacco rod **21** may function as a susceptor heated by the induction heater. Here, although not illustrated in the drawings, the tobacco rod **21** may further include an additional susceptor, in addition to the heat conductive material surrounding the tobacco rod **21**.

[0106] The filter rod **22** may include a cellulose acetate filter. Shapes of the filter rod **22** are not limited. For example, the filter rod **22** may include a cylinder-type rod or a tube-type rod having a hollow inside. Also, the filter rod **22** may include a recess-type rod. When the filter rod **22** includes a plurality of segments, at least one of the plurality of segments may have a different shape.

[0107] The first segment of the filter rod **22** may be a cellulous acetate filter. For example, the first segment may be a tube-type structure having a hollow inside. The first segment may prevent an internal material of the tobacco rod **21** from being pushed back when the heater **110** is inserted into the tobacco rod **21** and may also provide a cooling effect to aerosol. A diameter of the hollow included in the first segment may be an appropriate diameter within a range of 2 mm to 4.5 mm but is not limited thereto.

[0108] The length of the first segment may be an appropriate length within a range of 4 mm to 30 mm but is not limited thereto. For example, the length of the first segment may be 10 mm but is not limited thereto.

[0109] The second segment of the filter rod **22** cools the aerosol which is generated when the heater **110** heats the tobacco rod **21**. Therefore, the user may puff the aerosol which is cooled at an appropriate temperature.

[0110] The length or diameter of the second segment may be variously determined according to the shape of the stick **20**. For example, the length of the second segment may be an appropriate length within a range of 7 mm to 20 mm. Preferably, the length of the second segment may be about **14** mm but is not limited thereto.

[0111] The second segment may be manufactured by weaving a polymer fiber. In this case, a flavoring liquid may also be applied to the fiber formed of the polymer. Alternatively, the second segment may be manufactured by weaving together an additional fiber coated with a flavoring liquid and a fiber formed of a polymer. Alternatively, the second segment may be formed by a crimped polymer sheet.

[0112] For example, a polymer may be formed of a material selected from the group consisting of polyethylene (PE), polypropylene (PP), polyvinyl chloride (PVC), polyethylene terephthalate (PET), polylactic acid (PLA), cellulous acetate (CA), and aluminum coil.

[0113] As the second segment is formed by the woven polymer fiber or the crimped polymer sheet, the second segment may include a single channel or a plurality of channels extending in a longitudinal direction. Here, a channel refers to a passage through which a gas (e.g., air or aerosol) passes.

[0114] For example, the second segment formed of the crimped polymer sheet may be formed from a material having a thickness between about 5 μm and about 300 μm, for example, between about 10 μm and about 250 μm. Also, a total surface area of the second segment may be between about 300 mm2/mm and about 1000 mm2/mm. In addition, an aerosol cooling element may be formed from a material having a specific surface area between about 10 mm2/mg and about 100 mm2/mg.

[0115] The second segment may include a thread including a volatile flavor component. Here, the volatile flavor component may be menthol but is not limited thereto. For example, the thread may be filled with a sufficient amount of menthol to provide the second segment with menthol of 1.5 mg or more.

[0116] The third segment of the filter rod **22** may be a cellulous acetate filter. The length of the third segment may be an appropriate length within a range of 4 mm to 20 mm. For example, the length of the third segment may be about **12** mm but is not limited thereto.

[0117] The filter rod **22** may be manufactured to generate flavors. For example, a flavoring liquid may be injected onto the filter rod **22**. For example, an additional fiber coated with a flavoring liquid may be inserted into the filter rod **22**.

[0118] Also, the filter rod **22** may include at least one capsule **23**. Here, the capsule **23** may generate a flavor. The capsule **23** may generate an aerosol. For example, the capsule **23** may have a configuration in which a liquid including a flavoring material is wrapped with a film. The capsule **23** may have a spherical or cylindrical shape but is not limited thereto.

[0119] Referring to FIG. **6**, a stick **30** may further include a front-end plug **33**. The front-end plug **33** may be located on a side of a tobacco rod **31**, the side not facing a filter rod **32**. The front-end plug **33** may prevent the tobacco rod **31** from being detached and prevent liquefied aerosol from flowing into the aerosol generating device **10** from the tobacco rod **31**, during smoking.

[0120] The filter rod **32** may include a first segment **321** and a second segment **322**. The first segment **321** may correspond to the first segment of the filter rod **22** of FIG. **4**. The segment **322** may correspond to the third segment of the filter rod **22** of FIG. **4**.

[0121] A diameter and a total length of the stick **30** may correspond to the diameter and a total length of the stick **20** of FIG. **4**. For example, a length of the front-end plug **33** may be about 7 mm, a length of the tobacco rod **31** may be about 15 mm, a length of the first segment **321** may be about 12 mm, and a length of the second segment **322** may be about 14 mm, but embodiments are not limited thereto.

[0122] The stick **30** may be wrapped using at least one wrapper **35**. The wrapper **35** may have at least one hole through which external air may be introduced or internal air may be discharged. For example, the front-end plug **33** may be wrapped using a first wrapper **351**, the tobacco rod **31** may be wrapped using a second wrapper **352**, the first segment **321** may be wrapped using a third wrapper **353**, and the second segment **322** may be wrapped using a fourth wrapper **354**. Also, the entire stick **30** may be re-wrapped using a fifth wrapper **355**.

[0123] In addition, the fifth wrapper **355** may have at least one perforation **36** formed therein. For example, the perforation **36** may be formed in an area of the fifth wrapper **355** surrounding the tobacco rod **31** but is not limited thereto. For example, the perforation **36** may transfer heat formed by the heater **210** illustrated in FIG. **3** into the tobacco rod **31**.

[0124] Also, the second segment **322** may include at least one capsule **34**. Here, the capsule **34** may generate a flavor. The capsule **34** may generate an aerosol. For example, the capsule **34** may have a configuration in which a liquid including a flavoring material is wrapped with a film. The capsule **34** may have a spherical or cylindrical shape but is not limited thereto.

[0125] The first wrapper **351** may be formed by combining general filter wrapping paper with a metal foil such as an aluminum coil. For example, a total thickness of the first wrapper **351** may be within a range of 45 μm to 55 μm. For example, the total thickness of the first wrapper **351** may be 50.3 μm. Also, a thickness of the metal coil of the first wrapper **351** may be within a range 6 μm to 7 μm. For example, the thickness of the metal coil of the first wrapper **351** may be 6.3 μm. In addition, a basis weight of the first wrapper **351** may be within a range of 50 g/m2 to 55 g/m2. For example, the basis weight of the first wrapper **351** may be 53 g/m2.

[0126] The second wrapper **352** and the third wrapper **353** may be formed of general filter wrapping paper. For example, the second wrapper **352** and the third wrapper **353** may be porous wrapping paper or non-porous wrapping paper.

[0127] For example, porosity of the second wrapper **352** may be 35000 CU but is not limited thereto. Also, a thickness of the second wrapper **352** may be within a range of 70 μm to 80 μm. For example, the thickness of the second wrapper **352** may be 78 μm. A basis weight of the second wrapper **352** may be within a range of 20 g/m2 to 25 g/m2. For example, the basis weight of the second wrapper **352** may be 23.5 g/m2.

[0128] For example, porosity of the third wrapper **353** may be 24000 CU but is not limited thereto. Also, a thickness of the third wrapper **353** may be in a range of about 60 μm to about 70 μm. For example, the thickness of the third wrapper **353** may be 68 μm. A basis weight of the third wrapper **353** may be in a range of about 20 g/m2 to about 25 g/m2. For example, the basis weight of the third wrapper **353** may be 21 g/m2.

[0129] The fourth wrapper **354** may be formed of PLA laminated paper. Here, the PLA laminated paper refers to three-layer paper including a paper layer, a PLA layer, and a paper layer. For example, a thickness of the fourth wrapper **353** may be in a range of 100 μm to 1200 μm. For example, the thickness of the fourth wrapper **353** may be 110 μm. Also, a basis weight of the fourth wrapper **354** may be in a range of 80 g/m2 to 100 g/m2. For example, the basis weight of the fourth wrapper **354** may be 88 g/m2.

[0130] The fifth wrapper **355** may be formed of sterilized paper (MFW). Here, the sterilized paper (MFW) refers to paper which is particularly manufactured to improve tensile strength, water resistance, smoothness, and the like more than ordinary paper. For example, a basis weight of the fifth wrapper **355** may be in a range of 57 g/m2 to 63 g/m2. For example, the basis weight of the fifth wrapper **355** may be 60 g/m2. Also, a thickness of the fifth wrapper **355** may be in a range of 64 μm to 70 μm. For example, the thickness of the fifth wrapper **355** may be 67 μm.

[0131] The fifth wrapper **355** may include a preset material added thereto. An example of the material may include silicon, but it is not limited thereto. Silicon has characteristics such as heat resistance robust to temperature conditions, oxidation resistance, resistance to various chemicals, water repellency to water, and electrical insulation, etc. Besides silicon, any other materials having characteristics as described above may be applied to (or coated on) the fifth wrapper **355** without limitation.

[0132] The front-end plug **33** may be formed of cellulous acetate. For example, the front-end plug **33** may be formed by adding a plasticizer (e.g., triacetin) to cellulous acetate tow. Mono-denier of filaments constituting the cellulous acetate tow may be in a range of 1.0 to 10.0. For example, the mono-denier of filaments constituting the cellulous acetate tow may be within a range of 4.0 to 6.0. For example, the mono-denier of the filaments of the front-end plug **33** may be 5.0. Also, a cross-section of the filaments constituting the front-end plug **33** may be a Y shape. Total denier of the front-end plug **33** may be in a range of 20000 to 30000. For example, the total denier of the front-end plug **33** may be within a range of 25000 to 30000. For example, the total denier of the front-end plug **33** may be 28000.

[0133] Also, as needed, the front-end plug **33** may include at least one channel. A cross-sectional shape of the channel may be manufactured in various shapes.

[0134] The tobacco rod **31** may correspond to the tobacco rod **21** described above with reference to FIG. **4**. Therefore, hereinafter, the detailed description of the tobacco rod **31** will be omitted.

[0135] The first segment **321** may be formed of cellulous acetate. For example, the first segment **321** may be a tube-type structure having a hollow inside. The first segment **321** may be manufactured by adding a plasticizer (e.g., triacetin) to cellulous acetate tow. For example, mono-denier and total denier of the first segment **321** may be the same as the mono-denier and total denier of the front-end plug **33**.

[0136] The second segment **322** may be formed of cellulous acetate. Mono denier of filaments constituting the second segment **322** may be in a range of 1.0 to 10.0. For example, the mono denier of the filaments of the second segment **322** may be within a range of about 8.0 to about 10.0. For example, the mono denier of the filaments of the second segment **322** may be 9.0. Also, a cross-section of the filaments of the second segment **322** may be a Y shape. Total denier of the second segment **322** may be in a range of 20000 to 30000. For example, the total denier of the second segment **322** may be 25000.

[0137] Referring to FIG. **7**, the aerosol-generating device **10** may include a resistance detection sensor **150**, a temperature sensor **153**, a puff sensor **155**, a battery **16**, a power supply circuit **160**, and/or a first heater **210**.

[0138] According to an embodiment of the present disclosure, the resistance detection sensor **150**, the puff sensor **155**, the battery **16**, and/or the power supply circuit **160** may be disposed in the main body **100**. The first heater **210** may be disposed in the cartridge **200**.

[0139] When the main body **100** and the cartridge **200** are coupled to each other, the resistance detection sensor **150** of the main body **100** may be electrically connected to the first heater **210** of the cartridge **200**. For example, the resistance detection sensor **150** may be a current sensor for detecting current.

[0140] The power supply circuit **160**, which is disposed in the main body **100**, may supply power to the first heater **210** using the power stored in the battery **16**. In this case, the amount of power supplied from the power supply circuit **160** to the first heater **210** may be adjusted under the control of the controller **17**.

[0141] The power supply circuit **160** may include a converter that converts a voltage output from the battery **16**. For example, the power supply circuit **160** may include a buck-converter that steps down the voltage output from the battery **16**. In the present disclosure, the buck converter is described as an example of a configuration for converting a voltage, but embodiments are not limited thereto. For example, the power supply circuit **1210** may include a buck-boost converter, a Zener diode, and the like.

[0142] The power supply circuit **160** may include at least one switching element, which is operated under the control of the controller **17**. In this case, power may be supplied to the first heater **210** in response to operation of the switching element. For example, the switching element may be a bipolar junction transistor (BJT) or a field effect transistor (FET).

[0143] When the first heater **210** and the resistance detection sensor **150** are electrically connected to each other, current having the same magnitude may flow through the first heater **210** and the resistance detection sensor **150**. Here, the resistance Rs of the shunt resistor provided in the resistance detection sensor **150** may be a value that does not change with temperature.

[0144] The controller **17** may determine the voltage V1 applied to the first heater **210** and the resistance detection sensor **150** based on the power supplied from the power supply circuit **160** to the first heater **210** and the current flowing through the first heater **210** and the resistance detection sensor **150**. The controller **17** may calculate the voltage V2 applied to the shunt resistor of the resistance detection sensor **150** based on the current flowing through the shunt resistor and the resistance Rs of the shunt resistor. In this case, the controller **17** may calculate the voltage applied to the first heater **210** as the difference (V1−V2) between the voltage V1 applied to the first heater **210** and the resistance detection sensor **150** and the voltage V2 applied to the shunt resistor. In addition, the controller **17** may calculate the resistance Rh of the first heater **210** based on the voltage applied to the first heater **210** and the current flowing through the first heater **210**.

[0145] Accordingly, the controller **17** may determine the temperature of the first heater **210** in real time based on the current flowing through the first heater **210**, which is calculated by the resistance detection sensor **150**, even while the wick is being heated by the first heater **210**.

[0146] Meanwhile, the resistor of the first heater **210** may be a material having a temperature coefficient of resistance, and the resistance Rh of the first heater **210** may vary depending on changes in the temperature of the resistor. The controller **17** may calculate the temperature of the first heater **210** based on the temperature coefficient of resistance of the first heater **210**, the resistance Rh of the first heater **210**, and the resistance of the first heater **210** at a reference temperature using a calculation equation for calculating the temperature of the first heater **210**. Here, the calculation equation used to calculate the temperature of the first heater **210** may be expressed using the following Equation 1.

$$TCR = \frac{R1 - R0}{R0} \div (T1 - T0) \qquad \text{[Equation 1]}$$

[0147] In Equation 1 above, TCR represents the temperature coefficient of resistance of the first heater **210**, T1 represents the temperature of the first heater **210**, R1 represents the resistance of the first heater **210**, T0 represents the reference temperature, and R0 represents the resistance of the first heater **210** at the reference temperature. Here, T0 is 25° C., and R0 is the resistance of the first heater **210** at 25° C.

[0148] Although the current sensor is illustrated in this drawing as being connected in series to the first heater **210**, the present disclosure is not limited thereto. A temperature sensor disposed adjacent to the first heater **210** to detect the temperature of the first heater **210** or a voltage sensor for detecting the voltage applied to the first heater **210** may be provided as the resistance detection sensor **150**.

[0149] The temperature sensor **153** may output a signal corresponding to a temperature of gas flowing into the aerosol generating device **10**. For example, the temperature sensor **153** may be disposed in a flow path through which gas introduced into the aerosol generating device **10** flows. In this embodiment, the temperature sensor **153** is described as being implemented as a sensor configured to output a signal corresponding to the temperature of the gas flowing into the aerosol generating device **10**, but the present disclosure is not limited thereto. For example, the temperature sensor **153** may be a sensor disposed adjacent to battery **16** to detect a temperature of battery **16**.

[0150] The puff sensor **155** may output a signal corresponding to a puff. For example, the puff sensor **155** may output a signal corresponding to an internal pressure of the aerosol-generating device **10**. Here, the internal pressure of the aerosol-generating device **10** may correspond to the pressure in a flow path through which gas flows. In this embodiment, the puff sensor **155** is described as being implemented as a pressure sensor configured to output a signal corresponding to the internal pressure of the aerosol-generating device **10**, but the present disclosure is not limited thereto.

[0151] Meanwhile, according to one embodiment, the temperature sensor **153** and the puff sensor **155** may be implemented as one component.

[0152] The controller **17** may determine a puff based on the signal received from the puff sensor **155**. For example, the controller **17** may determine whether a puff occurs based on the sensing value of the signal from the puff sensor **155**. For example, the controller **17** may determine an intensity of a puff based on the sensing value of the signal from the puff sensor **155**. For example, the controller **17** may determine the time period during which a puff occurs (hereinafter referred to as a puff time period) based on the sensing value of the signal from the puff sensor **155**.

[0153] Upon determining that a puff has occurred, the controller **17** may control the aerosol-generating module **13**. For example, upon determining that a puff has occurred, the controller **17** may control the aerosol-generating module **13** such that power is supplied to the first heater **210** included in the aerosol-generating module **13**.

[0154] Upon determining that a puff has occurred, the controller **17** may update data stored in the memory **14**. For example, upon determining that a puff has occurred, the controller **17** may update the current number of puffs stored in the memory **14**. For example, upon determining that a puff has occurred, the controller **17** may update data on an intensity of the puff stored in the memory **14**.

[0155] FIGS. **8**A and **8**B are flowcharts showing an operation method of the aerosol-generating device according to an embodiment of the present disclosure.

[0156] Referring to FIG. **8**A, the aerosol generating device **10** may supply predetermined power to the first heater **210** in a first preheating period in operation S**801**. Here, the predetermined power may be power (hereinafter referred to as sensing power) supplied to the first heater **210** to detect the temperature of the first heater **210**.

[0157] In this embodiment, a period in which aerosol is generated through heating of the first heater **210** in response to detection of a puff through the puff sensor **155** may be referred to as a heating period. Meanwhile, a period in which a puff is not detected, for example, an interval from a point in time at which a puff ends to a point in time the puff is detected again may be referred to as a preheating period.

[0158] The aerosol generating device **10** may supply a predetermined minimum power (hereinafter, preheating power) to the first heater **210** based on a start of the preheating period. In this case, the preheating power may be less than the sensing power. For example, the aerosol generating device **10** may supply the preheating power to the first heater **210** from the start of the preheating period.

[0159] According to an embodiment, the aerosol generating device **10** may supply the sensing power greater than the preheating power to the first heater **210** based on a lapse of a predetermined time from the start of the preheating period. Here, the predetermined time may correspond to a time during which the liquid is supplied to the liquid delivery element (e.g., the wick) above a certain level. Meanwhile, a time during which the sensing power is supplied to the first heater **210** may be shorter than the predetermined time. For example, the predetermined time may be set to 3 seconds and the time for supplying the sensing power to 0.1 second.

[0160] The aerosol generating device **10** may determine whether the temperature of the first heater **210** corresponding to the supply of sensing power exceeds a predetermined first temperature in operation S**802**. Here, the first temperature may be a temperature (e.g., 210° C.) corresponding to a case in which the liquid is supplied to the liquid delivery clement (e.g., the wick) below a certain level. For example, in a case in which the liquid aerosol-generating substance is not exhausted, the temperature of the first heater **210** may be temporarily increased above the first temperature by the supply of the sensing power, when an amount of the aerosol-generating substance delivered to the liquid delivery element is temporarily decreased due to bubbles formed in the chamber.

[0161] The aerosol generating device **10** may stop preheating the first heater **210** based on the temperature of the first heater **210** exceeding the first temperature in operation S**803**. For example, the aerosol generating device **10** may interrupt the supply of the preheating power to the first heater **210**. Through this, the liquid may be more smoothly supplied to the liquid delivery element (e.g., the wick) before the heating period starts in a state in which the liquid aerosol-generating substance is not exhausted.

[0162] The aerosol generating device **10** may determine whether a puff is detected through the puff sensor **155** in operation S**804**. For example, the aerosol generating device **10** may determine that a puff has occurred based on the internal pressure of the aerosol generating device **10** being less than a reference pressure. For example, the aerosol generating device **10** may determine that a puff has occurred based on a change in the internal pressure of the aerosol generating device **10** being greater than or equal to a minimum change.

[0163] The aerosol generating device **10** may heat the first heater **210** based on the detection of the puff in operation S**805**. For example, the aerosol generating device **10** may supply power to the first heater **210** based on a preset temperature profile stored in the memory **14** so that the temperature of the first heater increases to a temperature for generating aerosol. In this case, the power supplied to the first heater **210** in the heating period (hereinafter referred to as heating power) may be greater than the sensing power.

[0164] According to an embodiment, predetermined power to be supplied to the first heater **210** in the heating period may vary according to the number of puffs, the time elapsed in the heating period, and the like. For example, power supplied to the first heater **210** while a puff is detected may decrease over time.

[0165] The aerosol generating device **10** may determine whether or not to end heating of the first heater **210** in operation S**806**. The aerosol generating device **10** may end heating of the first heater **210** when the puff ends. For example, the aerosol generating device **10** may determine that the puff has ended based on the internal pressure of the aerosol generating device **10** being less than the reference pressure. For example, the aerosol generating device **10** may determine that the puff has ended based on a gradient corresponding to a change in the internal pressure of the aerosol generating device **10** being greater than **0**.

[0166] The aerosol generating device **10** may supply the sensing power to the first heater **210** in the second preheating period in operation S**807**. For example, the aerosol generating device **10** may supply the sensing power to the first heater **210** based on a lapse of a predetermined time from the start of the second preheating period.

[0167] The aerosol generating device **10** may determine whether the temperature of the first heater **210** corresponding to the supply of the sensing power exceeds a predetermined second temperature in operation S**808**. Here, the second temperature may be a temperature higher than the first temperature. The second temperature may be a temperature (e.g., 240° C.) corresponding to a case in which the liquid contained in the liquid delivery clement (e.g., the wick) is below a minimum level due to exhaustion of the liquid.

[0168] The aerosol generating device **10** may determine that the liquid aerosol-generating substance is exhausted based on the temperature of the first heater **210** exceeding the second temperature in operation S**809**. In this case, the aerosol generating device **10** may interrupt the supply of power to the first heater **210** in response to exhaustion of the aerosol-generating substance. Meanwhile, the aerosol generating device **10** may keep interrupting the supply of power to the first heater **210** despite the puff sensor **155** detecting a puff. For example, the aerosol generating device **10** may cut off supply of power to the first heater **210** despite the puff sensor **155** detecting a puff.

[0169] Referring to FIG. **8**B, when the temperature of the first heater **210** is less than the first temperature in the first preheating period or when the temperature of the first heater **210** is less than the second temperature in the second preheating period, the aerosol generating device **10** may determine whether a puff is detected through the puff sensor **155** in operation S**810**. At this time, the aerosol generating device **10** may supply the preheating power to the first heater **210** until the puff is detected.

[0170] The aerosol generating device **10** may heat the first heater **210** based on the detection of the puff in operation S**811**.

[0171] According to an embodiment, based on the temperature of the first heater **210** detected in the first preheating period being equal to or less than the first temperature, first heating power may be supplied to the first heater **210** in a subsequent heating period. Meanwhile, based on the temperature of the first heater **210** detected in the first preheating period exceeding the first temperature, second heating power less than the first heating power may be supplied to the first heater **210** in a subsequent heating period. That is, the aerosol generating device **10** may supply relatively low power to the first heater **210** in the heating period when it is determined that the level at which the liquid is supplied to the liquid delivery clement (e.g., the wick) is less than a certain level. Through this, the liquid may be more smoothly supplied to the liquid delivery clement (e.g., the wick) until the sensing power is supplied to the first heater **210** in the second preheating period in a state in which the liquid aerosol-generating substance is not exhausted.

[0172] According to an embodiment, based on the temperature of the first heater **210** detected in the first preheating period being equal to or less than the first temperature, power may be supplied to the first heater **210** for a first heating time in a subsequent heating period. Meanwhile, based on the temperature of the first heater **210** detected in the first preheating period exceeding the first temperature, power may be supplied to the first heater **210** for a second heating time less than the first heating time in a subsequent heating period. That is, the aerosol generating device **10** may supply power to the first heater **210** for a relatively short time in the heating period when it is determined that the level at which the liquid is supplied to the liquid delivery clement (e.g., the wick) is less than a certain level. Through this, the liquid may be more smoothly supplied to the liquid delivery clement (e.g., the wick) until the sensing power is supplied to the first heater **210** in the second preheating period in a state in which the liquid aerosol-generating substance is not exhausted.

[0173] The aerosol generating device **10** may determine whether or not to end heating of the first heater **210** in operation S**812**. The aerosol generating device **10** may end heating of the first heater **210** when the puff ends. At this time, the aerosol generating device **10** may supply the preheating power to the first heater **210** in the first preheating period based on the end of the puff. In addition, the aerosol generating device **10** may redetermine whether the temperature of the first heater **210** corresponding to the supply of the sensing power in the first preheating period exceeds the first temperature.

[0174] Referring to FIGS. **9** and **10**, the preheating power P0 may be supplied to the first heater **210** until t1 included in the first preheating period. At this time, while the preheating power P0 is supplied to the first heater **210**, the temperature of the first heater **210** may be maintained at a target temperature T0 in the preheating period.

[0175] Meanwhile, the sensing power P1 may be supplied to the first heater **210** from t1 to t2. t1 may be a point in time when a predetermined time has elapsed from the start of the first preheating period. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 being less than the first temperature T1, the aerosol generating device **10** may determine that the liquid delivery clement (e.g., a wick) contains a certain level or more of liquid. In addition, the preheating power P0 may be continuously supplied to the first heater **210** even after t2.

[0176] The heating power P2 may be supplied to the first heater **210** based on the detection of the puff at t3. At this time, an aerosol may be generated due to the supply of the heating power P2 to the first heater **210**.

[0177] Meanwhile, from t4 when the puff ends, the preheating power P0 may be supplied to the first heater **210**. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 in the previous preheating period being equal to or less than the first temperature T1, the first preheating period may be started again from t4.

[0178] The sensing power P1 may be supplied to the first heater **210** from t5 to t6. t5 may be a point in time when a predetermined time has elapsed from t4. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 exceeding the first temperature T1, the aerosol generating device **10** may determine that the liquid is contained in the liquid delivery clement (e.g., the wick) below a certain level. In addition, the aerosol generating device **10** may interrupt the supply of the preheating power P0 to the first heater **210** based on the temperature of the first heater **210** exceeding the first temperature T1.

[0179] The heating power P2 may be supplied to the first heater **210** based on the detection of the puff at t7. In addition, the preheating power P0 may be supplied to the first heater **210** from the t8 when the puff ends. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 in the previous preheating period exceeding the first temperature T1, the second preheating period may be started from t8.

[0180] The sensing power P1 may be supplied to the first heater **210** from t9 to t10. t9 may be a point in time when a predetermined time has elapsed from t8. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 being less than the second temperature T2, the aerosol generating device **10** may determine that the liquid aerosol-generating substance is not exhausted. In addition, the preheating power P0 may be continuously supplied to the first heater **210** even after t10.

[0181] Referring to FIGS. **11** and **12**, the preheating power P0 may be supplied to the first heater **210** until t1 included in the first preheating period. At this time, while the preheating power P0 is supplied to the first heater **210**, the temperature of the first heater **210** may be maintained at a target temperature T0 in the preheating period.

[0182] Meanwhile, the sensing power P1 may be supplied to the first heater **210** from t1 to t2. t1 may be a point in time when a predetermined time has elapsed from the start of the first preheating period. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 exceeding the first temperature T1, the aerosol generating device **10** may determine that the liquid is contained in the liquid delivery element (e.g., the wick) below a certain level. In addition, the aerosol generating device **10** may interrupt the supply of the preheating power P0 to the first heater **210** based on the temperature of the first heater **210** exceeding the first temperature T1.

[0183] The heating power P2 may be supplied to the first heater **210** based on the detection of the puff at t3. In addition, from t4 when the puff ends, the preheating power P0 may be supplied to the first heater **210**. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 in the previous preheating period exceeding the first temperature T1, the second preheating period may be started from t4.

[0184] The sensing power P1 may be supplied to the first heater **210** from t5 to t6. t5 may be a point in time when a predetermined time has elapsed from t4. At this time, based on the temperature of the first heater **210** corresponding to the supply of the sensing power P1 exceeding the second temperature T2, the aerosol generating device **10** may determine that the liquid aerosol-generating substance is exhausted.

[0185] The aerosol generating device **10** may interrupt the supply of power to the first heater **210** from t6 in response to exhaustion of the liquid aerosol-generating substance.

[0186] According to one embodiment, in response to the exhaustion of the liquid aerosol-generating substance, a message about the exhaustion of the aerosol-generating substance may be output to the user. For example, the aerosol generating device **10** may output a screen corresponding to the exhaustion of the aerosol-generating substance through a display. For example, the aerosol generating device **10** may emit light corresponding to the exhaustion of the aerosol-generating substance through a light emitting diode (LED). For example, the aerosol generating device **10** may generate vibration corresponding to the exhaustion of the aerosol-generating substance through a motor.

[0187] Referring to FIG. **13**, according to an embodiment of the present disclosure, the insertion space, into which the stick **20** is inserted, may be defined in the upper end of the housing **101** of the aerosol generating device **10**.

[0188] The insertion space may be formed so as to be depressed to a predetermined depth toward the interior of the housing **101** so that the stick **20** is inserted at least partway thereinto. The depth of the insertion space may correspond to the length of the portion of the stick **20** that contains an aerosol-generating substance. For example, in the case in which the stick **20** shown in FIG. **5** is capable of being used in the aerosol-generating device **10**, the depth of the insertion space may correspond to the length of a tobacco rod **21** of the stick **20**.

[0189] Components such as battery **16**, the printed circuit board **1310**, and the heater may be disposed in the housing **101** of the aerosol-generating device **10**.

[0190] The components of the aerosol-generating device **10** may be mounted on one surface and/or the opposite surface of the printed circuit board **1310**. The components mounted on the printed circuit board **1310** may transmit or receive signals therebetween through a wiring layer of the printed circuit board **1310**. For example, at least one communication module included in the communication interface **11**, at least one sensor included in the sensor module **15**, the controller **17** and the like may be mounted on the printed circuit board **1310**.

[0191] The printed circuit board **1310** may be disposed adjacent to the battery **16**. For example, the printed circuit board **1310** may be disposed such that one surface thereof faces the battery **16**.

[0192] A display **1320** may be disposed on one side of the housing **101**. The display **1320** may display a screen in response to a signal transmitted from the controller **17**.

[0193] A power terminal **1330** may be disposed on one side of the housing **101** of the aerosol generating device **10**. The power terminal **1330** may be a wired terminal for wired communication such as USB.

[0194] A power supply circuit may be disposed between the battery **16** and the power terminal **1330**. The power supply circuit may transmit power supplied from the outside to the battery **16** through the power terminal **1330**. A power line **1335** for supplying power may be connected to the power terminal **1330**. For example, the power terminal **1330** may be coupled to a connector of the power line **1335**. The controller **17** may determine whether the power line **1335** is connected to the power terminal **1330**. For example, the controller **17** may determine whether the power line **1335** is connected to the power terminal **1330** based on a signal generated in response to connection between the power terminal **1330** and the power line **1335**.

[0195] A motor **1340**, which generates vibration for a haptic effect, may be disposed in the housing **101**. The motor **1340** may adjust the period and/or intensity of vibration based on a signal transmitted from the controller **17**.

[0196] The structure of the aerosol-generating device **10** is not limited to the structure shown in FIG. **13**. In some embodiments, the arrangement of the battery **16**, the printed circuit board **1310**, the display **1320**, the power terminal **1330**, the motor **1340**, and the like may vary.

[0197] According to one embodiment, when the second heater **115** for heating the stick **20** is provided, the aerosol generating device **10** may start an operation for generating aerosol based on insertion of the stick **20**. The second heater **115** may be referred to as a stick heater **115**. For example, the aerosol generating device **10** may preheat the second heater **115** based on detection of insertion of the stick **20** into the insertion space **130**. For example, the aerosol generating device **10** may supply preheating power to the first heater **210** based on completion of preheating of the second heater **115**.

[0198] According to an embodiment, the aerosol generating device **10** may detect a resistance of the first heater **210** based on the start of the operation for generating aerosol. At this time, the detected resistance of the first heater **210** may be determined as a resistance of the first heater **210** at a reference temperature used in a calculation formula for calculating the temperature of the first heater **210**. Meanwhile, the reference temperature used in the calculation formula for calculating the temperature of the first heater **210** may correspond to a temperature of gas detected through the temperature sensor **153** based on the start of the operation for generating aerosol. That is, the resistance of the first heater **210** and the temperature of gas, detected before the supply of power to the first heater **210** is started after the operation for generating aerosol is started, may be used in the calculation formula for calculating the temperature of the first heater **210**.

[0199] Meanwhile, when a user continuously uses a plurality of sticks **20**, the stick **20** may be reinserted into the insertion space **130** before the first heater **210** is sufficiently cooled. In addition, when the aerosol generating device **10** is stored in a low-temperature environment, the temperature of the first heater **210** may be relatively low at the time when the stick **20** is inserted into the insertion space **130** even if the user uses the aerosol generating device **10** in a room-temperature environment. In these cases, accurate detection of the reference temperature used in the calculation formula for calculating the temperature of the first heater **210** and/or the resistance of the first heater **210** at the reference temperature may be required.

[0200] According to one embodiment, when the aerosol generating device **10** includes the second heater **115** for heating the stick **20**, based on the supply of power to the second heater **115**, the reference temperature used in the calculation formula for calculating the temperature of the first heater **210** and/or the resistance of the first heater **210** at the reference temperature may be detected.

[0201] Referring to FIG. **14**, the aerosol generating device **10** may perform an operation of preheating the second heater **115** from time t0 to t1. **10** may be a point in time at which the insertion of the stick **20** into the insertion space **130** is detected, and t1 may corresponds to a target temperature Tpre for a temperature of the second heater **115**. At this time, while a time elapses from t0 to t1, the temperature of the second heater **210** may change to a temperature corresponding to the temperature of the environment in which the user uses the aerosol generating device **10**. Considering this point, the aerosol generating device **10** may determine the temperature of the gas detected through the temperature sensor **153** at t2 when a predetermined time elapses after the operation of preheating the second heater **115** starts as the reference temperature used in the calculation formula for calculating the temperature of the first heater **210**. In addition, the aerosol generating device **10** may determine the resistance of the second heater **115** detected at t2 when a predetermined time elapses after the operation of preheating the second heater **115** starts as the resistance of the first heater **210** at the reference temperature. Here, the predetermined time may be set corresponding to the t1 when the operation of preheating the second heater **115** ends. For example, t2 may be a time 2 seconds earlier than t1.

[0202] As described above, according to at least one of the embodiments of the present disclosure, it may be possible to determine whether a liquid aerosol-generating substance is smoothly supplied to a liquid delivery element based on a temperature of a heater **210** in a preheating period.

[0203] In addition, according to at least one of the embodiments of the present disclosure, it may be possible to smoothly supply a liquid aerosol-generating substance to a liquid delivery element when the liquid delivery element lacks the liquid aerosol-generating substance.

[0204] In addition, according to at least one of the embodiments of the present disclosure, it may be possible to accurately determine whether a liquid aerosol-generating substance is exhausted based on a temperature of a heater **210** in a preheating period.

[0205] Referring to FIGS. **1** to **14**, an aerosol-generating device **10** in accordance with one aspect of the present disclosure may include a chamber **220** configured to store a liquid, a heater **210** configured to heat the liquid, a resistance detection sensor **150** configured to output a signal corresponding to a resistance of the heater **210**, and a controller **17** configured to calculate a temperature of the heater **210** based on the resistance of the heater **210**. The controller **17** is configured to determine whether the temperature of the heater **210** exceeds a first temperature in response to supply of sensing power to the heater **210** in a first preheating period, determine whether the temperature of the heater **210** exceeds a second temperature greater than the first temperature in response to supply of the sensing power to the heater **210** in a second preheating period, based on the temperature of the heater **210** exceeding the first temperature, and determine that the liquid is exhausted based on the temperature of the heater **210** exceeding the second temperature.

[0206] In addition, in accordance with another aspect of the present disclosure, a heating period in which an aerosol is generated by heating the liquid is started in response to an end of the first preheating period, and the second preheating period is started in response to an end of the heating period.

[0207] In addition, in accordance with another aspect of the present disclosure, the controller **17** is configured to control so that preheating power less than the sensing power is supplied to the heater **210** based on a start of the first preheating period, and control so that the sensing power is supplied to the heater **210** when a predetermined time has elapsed from the start of the first preheating period.

[0208] In addition, in accordance with another aspect of the present disclosure, a time during which the sensing power is supplied to the heater **210** is less than the predetermined time.

[0209] In addition, in accordance with another aspect of the present disclosure, the controller **17** is configured to interrupt supply of power to the heater **210** until the first preheating period ends, based on the temperature of the heater **210** exceeding the first temperature.

[0210] In addition, in accordance with another aspect of the present disclosure,

[0211] In addition, in accordance with another aspect of the present disclosure, the sensing power is less than heating power supplied to the heater **210** for generating an aerosol by heating the liquid.

[0212] In addition, in accordance with another aspect of the present disclosure, the aerosol-generating device **10** may further comprise a housing **101** having an insertion space **130**, and a stick heater **115** configured to heat a stick **20** inserted into the insertion space **130**. The controller **17** is configured to start supply of power to the stick heater **115** based on insertion of the stick **20** into the insertion space **130**, and calculate the temperature of the heater **210** based on the resistance of the heater **210** detected at a specific point in time when a predetermined time has elapsed after the supply of power to the stick heater **115** is started.

[0213] In addition, in accordance with another aspect of the present disclosure, the aerosol-generating device **10** may further comprise a temperature sensor **153** configured to detect a temperature of gas flowing into the housing **101**. The controller **17** is configured to calculate the temperature of the heater **210** based on the temperature of gas detected at the specific point in time and the resistance of the heater **210**.

[0214] In addition, in accordance with another aspect of the present disclosure, the controller **17** is configured to control so that first heating power is supplied to the heater **210** in a heating period in which an aerosol is generated by heating the liquid, based on the temperature of the heater **210** being equal to or less than the first temperature, and control so that second heating power less than the first heating power is supplied to the heater **210** in the heating period, based on the temperature of the heater **210** exceeding the first temperature.

[0215] In addition, in accordance with another aspect of the present disclosure, the aerosol-generating device **10** may further comprise a motor **1340** configured to generate vibrations. The controller **17** is configured to control the motor **1340** to generate a vibration corresponding to exhaustion of the liquid, based on a determination that the liquid is exhausted.

[0216] Certain embodiments or other embodiments of the disclosure described above are not mutually exclusive or distinct from each other. Any or all elements of the embodiments of the disclosure described above may be combined with another or combined with each other in configuration or function.

[0217] For example, a configuration "A" described in one embodiment of the disclosure and the drawings and a configuration "B" described in another embodiment of the disclosure and the drawings may be combined with each other. Namely, although the combination between the configurations is not directly described, the combination is possible except in the case where it is described that the combination is impossible.

[0218] Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

1. An aerosol-generating device comprising:
a chamber shaped to store a liquid;
a heater configured to heat the liquid;
a resistance detection sensor configured to output a signal corresponding to a resistance of the heater; and
a controller configured to:
calculate a temperature of the heater based on the resistance of the heater;
determine whether the temperature of the heater exceeds a first temperature, based on a supply of sensing power to the heater during a first preheating period;
determine whether the temperature of the heater exceeds a second temperature greater than the first temperature, based on a supply of the sensing power to the heater during a second preheating period, and based on the

temperature of the heater having been determined to exceed the first temperature; and

determine that the liquid is exhausted based on the temperature of the heater exceeding the second temperature.

2. The aerosol-generating device according to claim **1**, wherein a heating period, during which an aerosol is generated by heating the liquid, is started in response to an end of the first preheating period,

wherein the second preheating period is started in response to an end of the heating period.

3. The aerosol-generating device according to claim **1**, wherein the controller is further configured to:

cause preheating power less than the sensing power to be supplied to the heater, based on a start of the first preheating period, and cause the sensing power to be supplied to the heater, based on a predetermined time having elapsed from the start of the first preheating period.

4. The aerosol-generating device according to claim **3**, wherein a time during which the sensing power is supplied to the heater is less than the predetermined time.

5. The aerosol-generating device according to claim **1**, wherein the controller is further configured to interrupt further supply of the sensing power to the heater until the first preheating period ends, based on the temperature of the heater exceeding the first temperature.

6. The aerosol-generating device according to claim **1**, wherein the sensing power is less than heating power supplied to the heater for generating an aerosol by heating the liquid.

7. The aerosol-generating device according to claim **1**, further comprising:

a housing shaped to define an insertion space; and

a stick heater configured to heat a stick positioned in the insertion space,

wherein the controller is further configured to:

start supply of power to the stick heater, based on the stick being positioned in the insertion space, and

calculate the temperature of the heater, based on the resistance of the heater detected at a specific point in time after a predetermined time that the supply of power to the stick heater was started.

8. The aerosol-generating device according to claim **7**, further comprising a temperature sensor configured to detect a temperature of gas flowing into the housing,

wherein the controller is further configured to calculate the temperature of the heater, based on the temperature of gas detected at the specific point in time and the resistance of the heater.

9. The aerosol-generating device according to claim **1**, wherein the controller is further configured to:

cause first heating power to be supplied to the heater during a heating period in which an aerosol is generated by heating the liquid, based on the temperature of the heater being equal to or less than the first temperature, and

cause second heating power less than the first heating power to be supplied to the heater during the heating period, based on the temperature of the heater exceeding the first temperature.

10. The aerosol-generating device according to claim **1**, further comprising a motor configured to generate vibrations,

wherein the controller is further configured to control the motor to generate a vibration corresponding to exhaustion of the liquid, based on the determination that the liquid is exhausted.

11. An aerosol-generating device comprising:

a chamber shaped to store a liquid aerosol-generating substance;

a heater configured to heat the liquid aerosol-generating substance;

a resistance detection sensor configured to output a signal corresponding to a resistance of the heater; and

a controller configured to:

calculate a temperature of the heater based on the resistance of the heater;

determine that the temperature of the heater exceeds a first temperature, based on a supply of sensing power to the heater during a first preheating period;

determine, after the temperature of the heater has been determined to exceed the first temperature, that the temperature of the heater exceeds a second temperature greater than the first temperature, based on a supply of the sensing power to the heater during a second preheating period that occurs after the first preheating period; and

determine that the liquid aerosol-generating substance is exhausted based on the temperature of the heater exceeding the second temperature.

12. The aerosol-generating device according to claim **11**, further comprising:

a housing shaped to define an insertion space; and

a stick heater configured to heat a stick located in the insertion space, wherein the controller is further configured to:

start supply of power to the stick heater, after the stick is located in the insertion space, and

calculate the temperature of the heater, based on the resistance of the heater detected after a predetermined time that the supply of power to the stick heater was started.

13. The aerosol-generating device according to claim **12**, further comprising a temperature sensor configured to detect a temperature of gas flowing in the housing,

wherein the controller is further configured to calculate the temperature of the heater, based on the temperature of gas and the resistance of the heater.

14. The aerosol-generating device according to claim **11**, wherein the controller is further configured to:

cause first heating power to be supplied to the heater during a heating period in which an aerosol is generated by heating the liquid aerosol-generating substance, based on the temperature of the heater being equal to or less than the first temperature, and

cause second heating power less than the first heating power to be supplied to the heater during the heating period, based on the temperature of the heater exceeding the first temperature.

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